BIOC0001 – Proteins and Enzymes

Properties of Enzymes

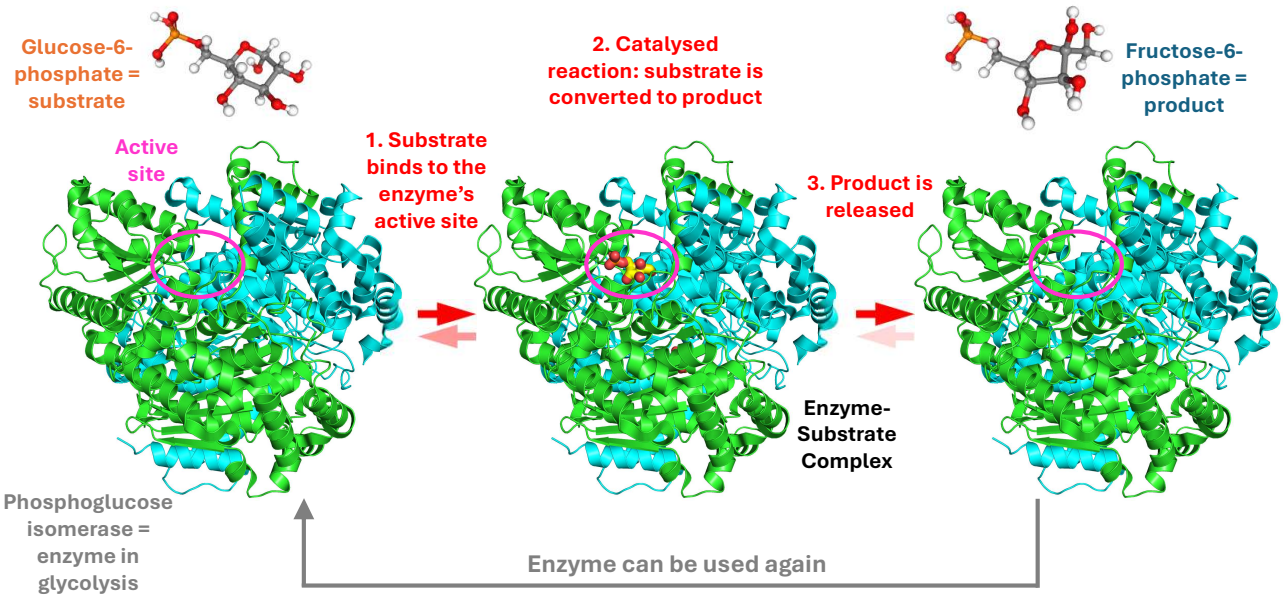
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Enzymes are biological catalysts that speed up biochemical reactions in living organisms. But enzyme technology is also used every day in households and industries. Examples include:

- **Rennet** contains the proteases chymosin and pepsin. In **cheese production**, rennet acts on casein, a major protein in milk, causing the milk to coagulate and separate into solid curds and liquid whey. Historically, rennet was extracted from the stomach lining of young mammals, particularly calves. Today, microbial and fermentation-produced alternatives are widely used.
- **Alpha-amylase** breaks down starch into smaller carbohydrates. In **textile manufacturing**, it is used to remove starch-based sizing agents from woven cotton. Starch is applied to yarns to improve their strength during weaving. Removing them helps prepare the fabric for subsequent processes, such as dyeing.
- **Bromelain** is a group of proteases found in pineapple tissues, including the stem and fruit. In the food industry, bromelain is commonly used as a **meat tenderiser**. It breaks down proteins in muscle tissue, making the meat softer.
- **Lipases** break down fats and oils. They are commonly added to **laundry detergents** to help remove greasy stains, such as cooking oil, butter, grease, and sunscreen. Many lipases used in laundry detergents are designed to work effectively at relatively low washing temperatures, helping to remove stains while reducing energy consumption.

In this section, we describe some properties of enzymes.

Enzyme-catalysed Reactions



PDB: 1HOX, rabbit phosphoglucose isomerase (homodimer) complexed with fructose-6-phosphate

Enzymes are **biological catalysts** that **increase the speed of a chemical reaction without themselves undergoing any permanent chemical change**. They are not used up in the reaction and they do not appear as reaction products.

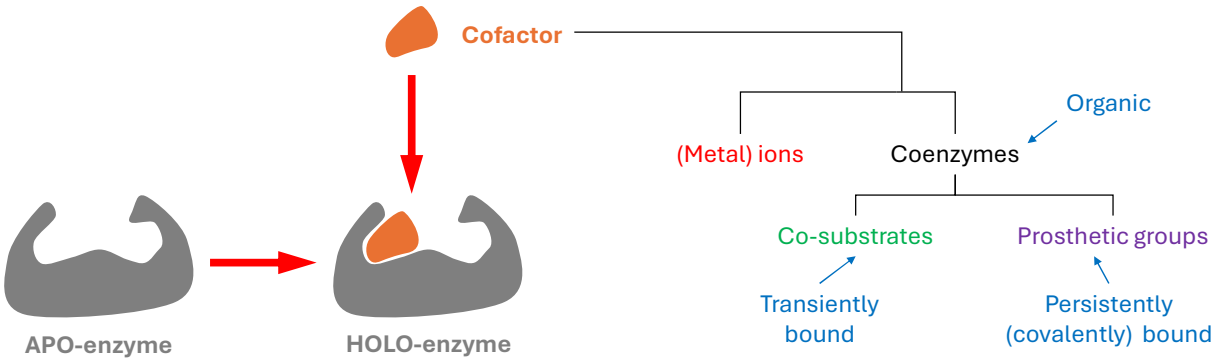
An enzymatic reaction can be described in three main steps:

- **Substrate binding:** The substrate binds to the enzyme's **active site** through non-covalent attractive forces. An **enzyme-substrate complex** is formed.
- **Catalysis:** The chemical reaction occurs. Covalent bonds within the substrate are broken and/or new bonds are formed, producing an **enzyme-product complex**.
- **Product release:** The product is released, and the enzyme is free to bind another substrate. This cycle can repeat many times.

The enzymatic reaction occurs in the active site, which contains residues involved in substrate recognition and binding and residues involved in catalysis.

Of note, the figure shown is just one example, where a single substrate is converted into a single product. In other reactions, there may be multiple substrates and/or multiple products (*e.g.* two substrates forming one product).

Cofactors



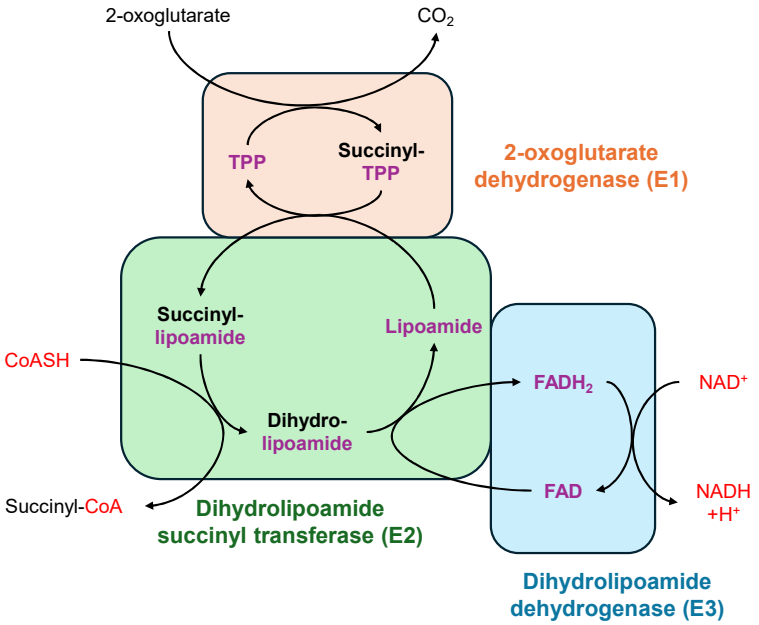
Cofactor examples: Fe^{2+} , Fe^{3+} , Mn^{2+} , Mg^{2} , K^{+} , thiamine pyrophosphate (TPP) (derived from vitamin B1 = thiamine) , flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN) (derived from vitamin B2 = riboflavin) , nicotinamide adenine dinucleotide (NAD) (derived from vitamin B3 = niacin) , Coenzyme A (CoASH) (derived from vitamin B5 = pantothenic acid) …

Many enzymes require an additional chemical component called a **cofactor.** These are either inorganic **ions** or organic molecules called **coenzymes**.

Coenzymes act as transient carriers of specific functional groups. Most are derived from vitamins, organic nutrients required in small amounts in the diet. **Prosthetic groups** refer to covalently bound coenzymes, while **co-substrates** refer to loosely bound coenzymes that are released in the same way as substrates and products.

The term **holoenzyme** refers to a catalytically active enzyme that consists of both **apoenzyme** (enzyme without its cofactor(s)) and cofactor(s).

Example: 2-oxoglutarate dehydrogenase complex



2-oxoglutarate + NAD^+ + CoASH

⇕

Succinyl-CoA + NADH + CO_2

Metal ion:
Mg^2

Prosthetic groups:
Thiamine pyrophosphate (TPP)
Flavin adenine dinucleotide (FAD)
Lipoamide

Co-substrates:
Nicotinamide adenine dinucleotide (NAD)
Coenzyme A (CoASH)

For example, the 2-oxoglutarate (or α-ketoglutarate) dehydrogenase complex from the citric acid cycle – which consists of three separate enzymes connected by linkers that catalyse the oxidative decarboxylation of 2-oxoglutarate to succinyl CoA – uses five coenzymes and one metal ion:

- The prosthetic groups thiamine pyrophosphate (TPP), lipoamide and flavin adenine dinucleotide (FAD)
- The co-substrates nicotinamide adenine dinucleotide (NAD^+) and coenzyme A (CoA)
- The metal ion Mg^{2+}

We need to obtain these coenzymes as the vitamins thiamine (B1), riboflavin (B2), niacin (B3), and pantothenic acid (B5, for CoASH) in our diet.

Specificity

The 8 enzymes of the citric acid cycle

Each enzyme will catalyse only one reaction. Cofactors serve many enzymes.

1. Citrate synthase
2. Aconitase
3. Isocitrate dehydrogenase
4. 2-oxoglutarate (or α-ketoglutarate) dehydrogenase complex
5. Succinyl-CoA synthetase
6. Succinate dehydrogenase
7. Fumarase
8. Malate dehydrogenase

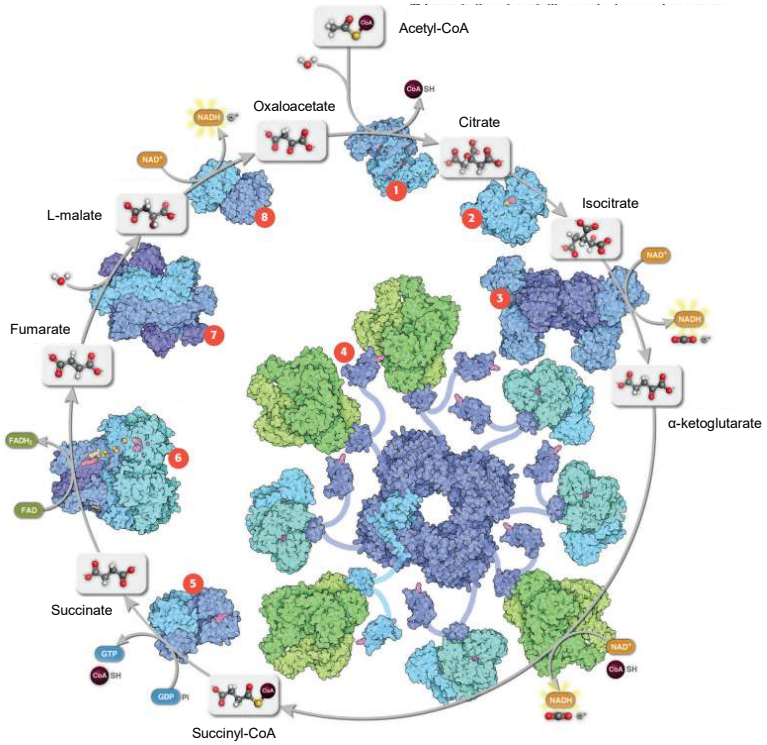


Figure: https://cdn.rcsb.org/pdb101/learn/resources/flyers/citric_acid_cycle.pdf

Most enzymes are **proteins** (with the exception of some catalytic RNAs called ribozymes).

One of the main properties that makes enzymes so valuable in research and diagnostics is their **specificity**: they catalyse particular reactions rather than many different ones. Some enzymes show **absolute specificity**, meaning they catalyse only one specific reaction. For example, in the citric acid cycle, each enzyme acts on a specific substrate. Aconitase, for instance, converts citrate to isocitrate. Other enzymes show **linkage specificity**, acting on a particular type of chemical bond, or **group specificity**, acting on a specific functional group.

The activity and specificity of an enzyme depends on its **unique three-dimensional structure**, which determines how it recognizes and interacts with its specific substrate(s) through a precise fit in both shape and charge, known as **geometric and electrostatic complementarity**.

Though enzymes exhibit great degrees of specificity, cofactors may serve many apoenzymes! For example, **nicotinamide adenine dinucleotide (NAD)** is a coenzyme for a great number of **dehydrogenase reactions** in the citric acid cycle.

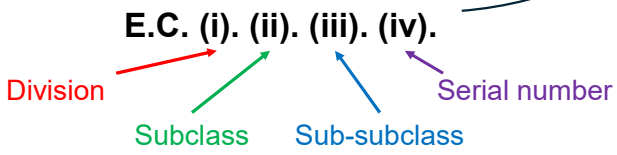
Names and Classification

Reaction: ATP + D-hexose ⇄ ADP + D-hexose-6-phosphate

E.C. number *(classification)*: 2.7.1.1.

Systematic name: ATP : D-hexose 6-phosphotransferase

Trivial name: hexokinase



See https://iubmb.qmul.ac.uk/enzyme/EC2/7/1/1.html

An enzyme may have up to three different names. All enzymes have a **four-digit E.C. number** based on the type of reaction they catalyze and a **systematic name** according to the reaction they catalyze. Some enzymes have a **working or trivial name** too.

The **Enzyme Commission** devised a system for the **classification of enzymes**. The code numbers, prefixed by EC, contain four elements separated by points, with the following meaning:

- the first number shows to which of the seven main divisions (classes) the enzyme belongs,
- the second figure indicates the subclass,
- the third figure gives the sub-subclass,
- the fourth figure is the serial number of the enzyme in its sub-subclass.

The enzyme classifications can be found here: https://iubmb.qmul.ac.uk/enzyme/.

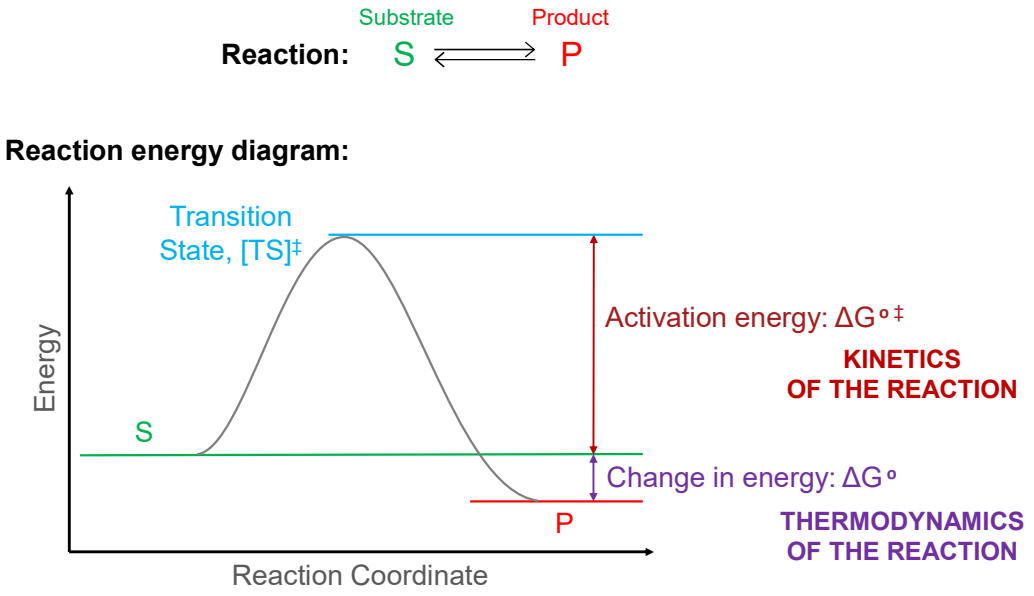
Classes

Division	Reaction	Example
1. Oxidoreductases	$A_{red} + B_{ox} \rightleftarrows A_{ox} + B_{red}$	Catalase
2. Transferases	$AB + C \rightleftarrows A + BC$	Glutathione-S-transferase
3. Hydrolases	$AB + H_2O \rightleftarrows A\text{-}H + B\text{-}OH$	Carboxypeptidase A
4. Lyases	$A{=}B + X\text{-}Y \rightleftarrows A(X)\text{-}B(Y)$	Pyruvate decarboxylase
5. Isomerases	$A \rightleftarrows B$	Malate isomerase
6. Ligases	$A + B + XTP \rightleftarrows AB + XDP + P_i$	Pyruvate carboxylase
7. Translocases		Cytochrome-c oxidase

The **seven main divisions** are:

1. **Oxidoreductases** are involved in oxidation / reduction reactions, *i.e.* they transfer electrons. They use oxygen, H_2O_2 or NAD^+ … as an electron acceptor.
2. **Transferases** transfer a functional group, *e.g.* a phosphate or an amino group.
3. **Hydrolases** catalyse the hydrolytic cleavage of C-O, C-N, C-C and some other bonds, *i.e.* they transfer water.
4. **Lyases** add or remove water, carbon dioxide or ammonia to break or form double bonds
5. **Isomerases** catalyse geometric or structural changes within one molecule.
6. **Ligases** join two molecules coupled with the hydrolysis of a diphosphate bond in ATP or a similar triphosphate.
7. **Translocases** *(created in 2018)* translocate hydrogen ions, inorganic cations and anions, amino acids, carbohydrates, or other compounds (usually across cell membranes).

Thermodynamics and Kinetics



A simple substrate-to-product net reaction is shown, where S and P represent the substrate and product.

This reaction can be described by a **reaction energy diagram**. The Y-axis is the **free energy** of a substrate molecule undergoing the reaction under defined conditions, *e.g.* standard conditions (298 K; 1 atm; all solutes at 1 M). The X-axis is the **reaction coordinate**, *i.e.* the extent to which a substrate molecule has changed into a product molecule.

In the example shown, the energy level of the product is lower than that of the substrate. This tells us that the **change in free energy for the reaction ($\Delta G^\circ = G^\circ_P - G^\circ_S$)** is negative. This reaction favours product formation.
The change in free energy is what determines the thermodynamics of a reaction, the degree to which a reaction is favoured or disfavoured.

An 'energy barrier' must be overcome to get to the product. At the very top of the energy barrier, the reaction is at its **transition state ($[TS]^\ddagger$)**, which is the point at which the bonds are in the process of breaking and forming. The height of this energy barrier is called the **activation energy ($\Delta G^{\circ\ddagger} = G^{\circ\ddagger} - G^\circ_S$)**. The higher the activation energy, the slower the reaction.
The activation energy is what determines the kinetics of a reaction, the rate at which a reaction occurs.

Thermodynamics and Equilibrium

$$A + B \rightleftharpoons C + D \qquad \textit{eq 1}$$

$$K_{eq} = \frac{[C]_{eq}[D]_{eq}}{[A]_{eq}[B]_{eq}} \quad \text{with eq} = \text{equilibrium} \qquad \textit{eq 2}$$

Gibb's: $$\Delta G = \Delta G^{o} + RT \ln \frac{[C]_i[D]_i}{[A]_i[B]_i} \quad \text{with i} = \text{initial} \qquad \textit{eq 3}$$

At equilibrium: $0 = \Delta G^{o} + RT \ln \frac{[C]_{eq}[D]_{eq}}{[A]_{eq}[B]_{eq}}$ and $\Delta G^{o} = -RT \ln K_{eq}$ *eq 4 and 5*

When $K_{eq} > 1$, ΔG^0 is negative; products are favoured over reactants.

When $K_{eq} < 1$, ΔG^0 is positive, reactants are favoured over products.

When $K_{eq} = 1$, $\Delta G^0 = 0$.

A reaction's thermodynamics is the degree to which the reaction is "favourable"— in other words, how much energy is released into the surroundings (or taken from the surroundings) as the reactants are transformed into products. The quantity of energy released or absorbed by a system as the reactants become products is called **change in free energy for the reaction ($\Delta G = G_P - G_S$)**.

- A **negative ΔG** indicates a **favourable**, or "downhill", reaction as it describes a process that decreases the system's free energy.
- A **positive ΔG** describes a process that increases the system's free energy. Such an "uphill" process is **unfavourable** and does not take place spontaneously.

ΔG in a chemical reaction is dependent on many factors including the concentrations of reactants and products, the temperature of the reaction… This means that ΔG is constantly changing for a system that is not at equilibrium. When **ΔG = 0**, the system is at **equilibrium**, **the rate of formation of products equals the rate at which products are converted to reactants**. There is no net change in the concentrations of reactants and products, so no driving force remains, and the system cannot do any work, *i.e.* ΔG = 0. The **equilibrium constant K_{eq}** is defined by *(eq 1 and 2)*, where $[]_{eq}$ represents the equilibrium concentration of any molecular species.

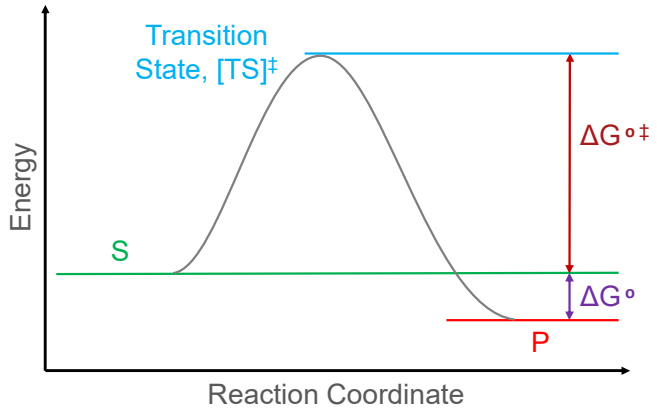
Gibbs showed that *(eq 3)*, where $[]_i$ represent the initial concentration of any molecular species, *R* is the gas constant, *T* is the temperature and **ΔG^0** is **the change in free energy for the *hypothetical* reaction under standard conditions (temperature, 298 K; pressure, 1 atm; all solutes at 1 M)**.

Under equilibrium conditions, we can substitute $\Delta G = 0$ and $[]_i = []_{eq}$, and we obtain *(eq 4 and 5)*.

- When **$K_{eq} > 1$, ΔG^0 is negative; products are favoured over reactants.**
- When **$K_{eq} < 1$, ΔG^0 is positive, reactants are favoured over products.**
- When **$K_{eq} = 1$, $\Delta G^0 = 0$.**

Because K_{eq} under standard conditions is measurable, we can determine ΔG^0. Knowing the value of ΔG^0, we can calculate the value of ΔG for the reaction away from equilibrium.

Kinetics and Rate



$$S \longrightarrow [TS]^{\ddagger} \longrightarrow P \qquad \textit{eq 1}$$

$$k = A \exp\left(\frac{-\Delta G^{\ddagger}}{R\,T}\right) \qquad \textit{eq 2}$$

$$V = -\frac{d[S]}{dx} = \frac{d[P]}{dx} = k\,[S] \qquad \textit{eq 3}$$

Rate constant

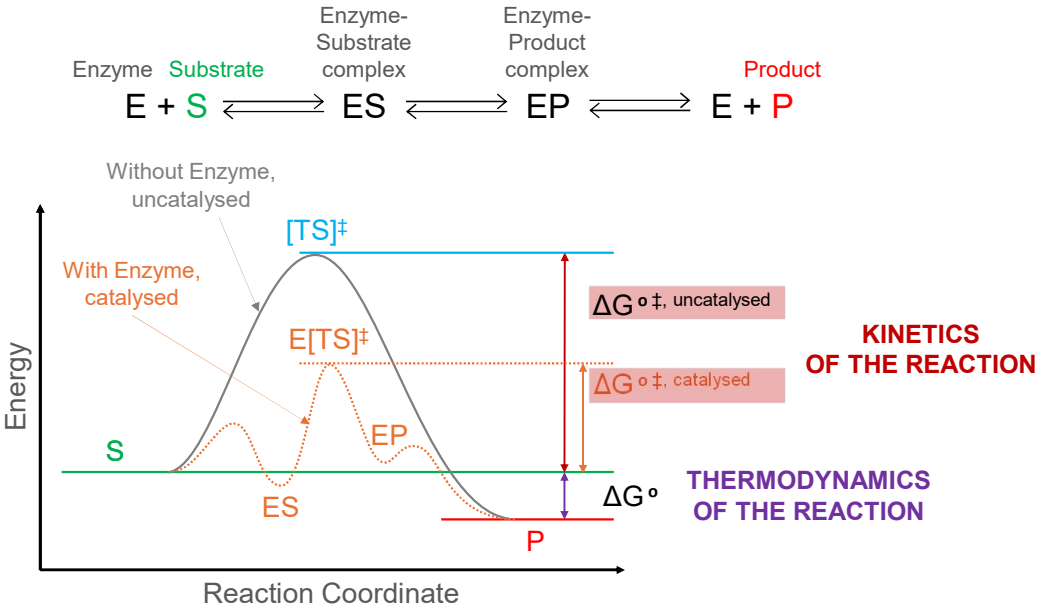
For the reactants to become products, an 'energy barrier' must be overcome. At the very top of the energy barrier, the reaction is at its **transition state ([TS]‡)**, which is the point at which the bonds are in the process of breaking and forming. Because the energy of the transition state is at a maximum, the system can spontaneously become products (or revert to reactants) without requiring the infusion of any additional energy.

The height of this energy barrier is called the **activation energy ($\Delta G^{\ddagger} = G^{\ddagger} - G_S$). $\Delta G^{\ddagger}$ is always positive**. Since chemical reactions proceed to product by going through the transition state *(eq 1)*, the rate of a chemical reaction is directly proportional to the concentration of the transition state at any given moment. T**he higher the $\Delta G^{\ddagger}$, the harder it is to reach the transition state, the lower the concentration of the transition state, and the slower the reaction**.

The rate of the reaction shown in *(eq 1)* depends on the substrate concentration and the **rate constant *k***, which is related to $\Delta G^{\ddagger}$, shown in *(eq 2)*, where A is a constant called the pre-exponential factor, *R* is the gas constant, and *T* is the temperature. The relationship between the rate constant *k* and the activation energy $\Delta G^{\ddagger}$ is inverse and exponential. Larger values of $\Delta G^{\ddagger}$ result in smaller rate constants.

In the case of the simplest **first-order reaction**, *e.g.* the uncatalysed S → P reaction *(eq 1)*, the rate of the reaction can be written as *(eq 3)*. This is the **rate equation**. In this case, the rate of the reaction is linearly proportional to the concentration of S and the units of the first-order rate constant *k* are s^{-1}.

How Do Enzymes Work?



A simple enzymatic reaction is shown, where E, S, and P represent the enzyme, substrate, and product; ES and EP are the enzyme-substrate and enzyme-product complexes.

Enzymes accelerate chemical reactions by lowering the activation energy of the reaction ($\Delta G^{o\ddagger}$) (reducing the energy barrier to overcome).

Because an enzyme is not changed by a chemical reaction, **the change in free energy for the reaction (ΔG^o) is never changed by an enzyme**.

The ES and EP complexes occupy valleys in the reaction coordinate diagram and can be considered **reaction intermediates**.

<u>To conclude: enzymes can alter the reaction kinetics (rate), but not the reaction thermodynamics (equilibrium)!</u>

Rate Enhancement

$$Rate\ enhancement = \frac{k_{cat}}{k_{uncat}} = \exp\left(\frac{\Delta G^{\ddagger,uncatalysed} - \Delta G^{\ddagger,catalysed}}{R\,T}\right)$$

Enzyme	k_{uncat} Nonenzymatic Reaction Rate (s^{-1})	k_{cat} Enzymatic Reaction Rate (s^{-1})	$\frac{k_{cat}}{k_{uncat}}$ Rate Enhancement
Carbonic anhydrase	1.3×10^{-1}	1×10^{6}	7.7×10^{6}
Chorismate mutase	2.6×10^{-5}	50	1.9×10^{6}
Triose phosphate isomerase	4.3×10^{-6}	4300	1.0×10^{9}
Carboxypeptidase A	3.0×10^{-9}	578	1.9×10^{11}
AMP nucleosidase	1.0×10^{-11}	60	6.0×10^{12}
Staphylococcal nuclease	1.7×10^{-13}	95	5.6×10^{14}

Table: Voet & Voet, Biochemistry, John Wiley and Sons

Comparing k_{cat} (reaction rate in the presence of enzyme) to k_{uncat} (reaction rate in the absence of enzyme) gives us the **rate enhancement** of the reaction. It measures how much the enzyme decreases the activation energy, $\Delta G^{\ddagger}$, for a given reaction.

The rates of enzymatically catalysed reactions are typically 10^6 to 10^{14} times greater than those of the corresponding uncatalysed reactions.

Substrate Binding and Catalysis

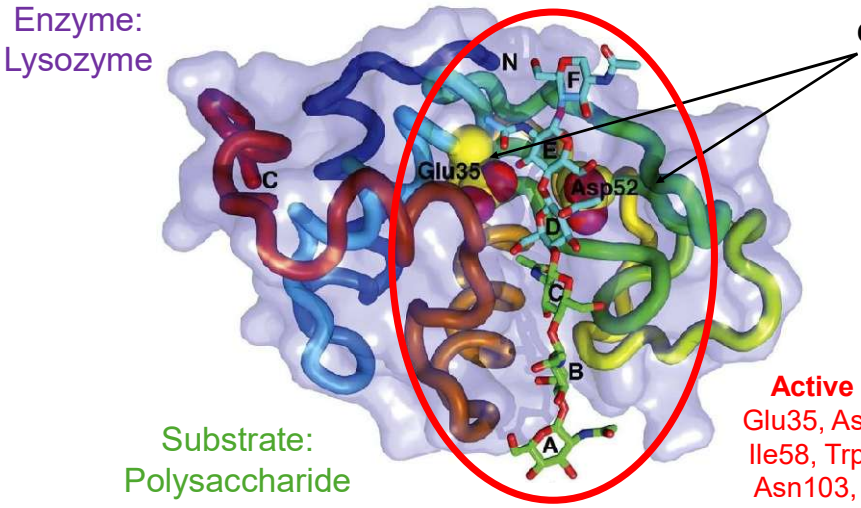


Figure: Voet & Voet, Biochemistry, John Wiley and Sons

Enzymes can change the rate of a reaction by 6 to 14 orders of magnitude through decreasing the activation energy ($\Delta G^{\ddagger}$). How do they do this?

A significant part of the energy used for enzymatic rate enhancements comes from **enzyme-substrate binding**. The binding involves many **weak, non-covalent interactions** (*i.e.* hydrogen bonds, salt bridges, hydrophobic, and van der Waals interactions). Importantly, enzymes are designed to bind the transition state of the reaction more tightly than the substrate itself. This tight binding releases energy that helps lower the activation energy. These interactions help the reaction in several ways:

- **Reducing entropy**: The enzyme holds the substrate in place, limiting its movement
- **Desolvation**: Water molecules are removed from the substrate, exposing reactive groups
- **Proper alignment**: The substrate is positioned exactly where it needs to be for the reaction to occur

Enzyme active sites contain **catalytic residues** and sometimes cofactors that directly participate in the making and breaking of bonds during catalysis.

Example: Lysozyme hydrolyses glycosidic bonds between the monosaccharide units of the cell walls of bacteria and fungi. It is a single chain of 129 residues with an α+β fold. The enzyme has a cleft forming the active site. Six monosaccharide units – from A to F, shown as sticks – bind in the active site. Many active site residues are involved in its binding. The side chains of the catalytic residues, Glu35 and Asp52, are shown as spheres. These catalyse the hydrolysis of the glycosidic bond between monosaccharide units D and E.

Models for Substrate Binding

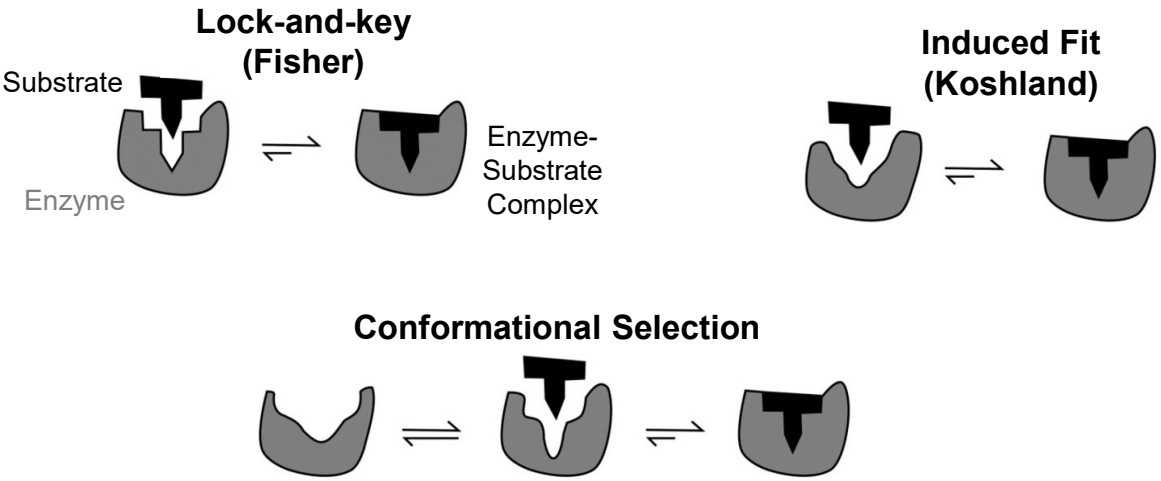


Figure: Insights into Protein–Ligand Interactions: Mechanisms, Models, and Methods

Different models have been proposed to explain enzyme-substrate binding:

- The "**lock-and-key**" model (by Emil Fischer in 1890) assumes that both the protein and the ligand are rigid, and their binding interfaces are perfectly matched. As a result, only the correctly sized ligand (the key) can insert into the binding pocket of the protein (the lock).
- The "**induced fit**" model (by Daniel Koshland in 1958) assumes some flexibility. (Minor) conformational changes in the protein and / or the ligand allow the complementary groups to interact fully.
- The "**conformational selection**" model postulates that the native state of a protein does not exist as a single, rigid conformation but rather as a vast ensemble of conformational substates that coexist in equilibrium with different substate distributions The ligand can bind selectively to the most suitable conformational substate, ultimately shifting the equilibrium towards this substate.

Catalytic Mechanisms

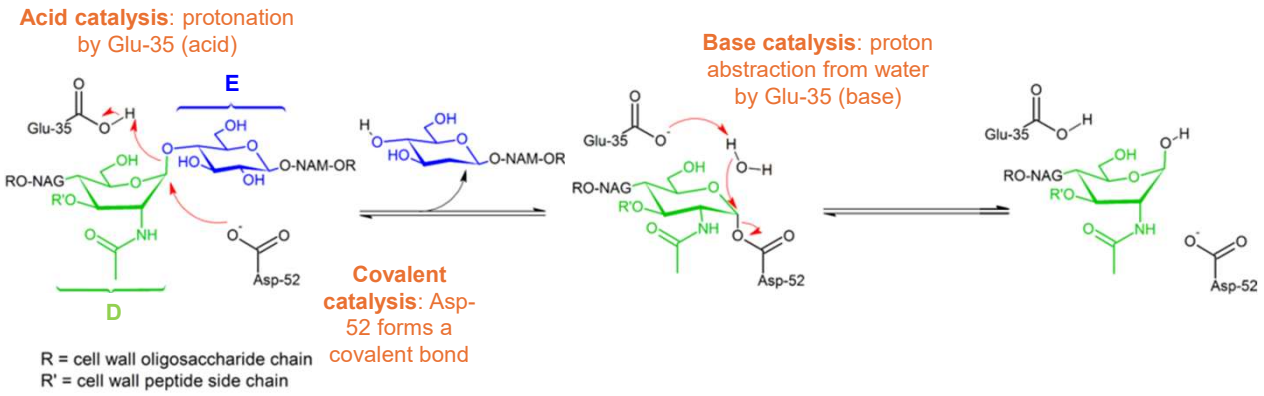


Figure: Mechanism_of_Lysozyme_action.png (1024×276)

Covalent interactions between the substrate and the enzyme's functional groups (**covalent catalysis**), proton transfers to and from the enzyme's functional groups (**general acid-base catalysis**), and interactions with metal ions (**metal ion catalysis**) can provide a new, lower-energy reaction path.

We will focus on acid-base catalysis (remember the amino acid titration curves!?):

- General acid catalysis involves partial proton transfer from a donor that lowers the free energy of the transition state
- General base catalysis involves partial proton abstraction from an acceptor that lowers the free energy of the transition state

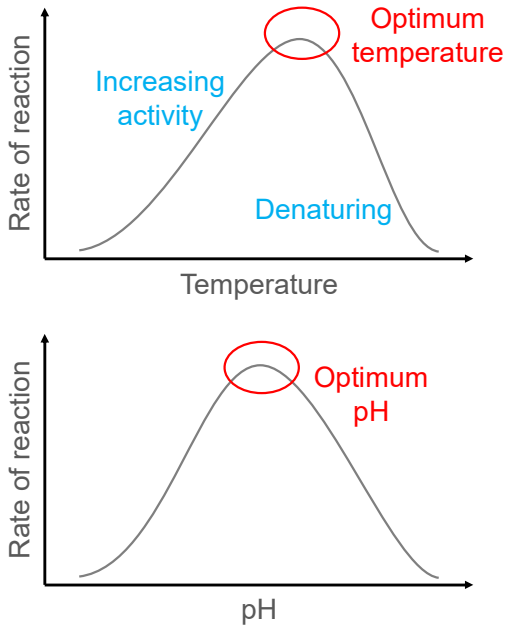
Lysozyme makes use of covalent catalysis and acid-base catalysis:

- *Step 1*: Asp52 acts as a covalent catalyst, directly displacing monosaccharide unit E, forming a glycosyl enzyme intermediate. Glu35 acts as a general acid catalyst and protonates monosaccharide unit E to facilitate its departure.
- *Step 2*: Glu35 acts as a general base catalyst by abstracting a proton from water and forming a hydroxyl ion. This attacks the glycosyl enzyme intermediate, displacing Asp52 and generating the product.

Factors affecting enzyme activity

- Temperature
- pH
- [Enzyme]
- [Substrate]
- [Inhibitor]
- [Activator]

→ See the online lab!



Several factors affect the rate at which enzymatic reactions proceed: enzyme concentration, substrate concentration, the presence of any inhibitors or activators, temperature, and pH.

Enzymes are affected by changes in temperature. Like most chemical reactions, the rate of an enzyme-catalysed reaction increases as the temperature is raised. The rate increases to a maximum level, then abruptly declines with further increase of temperature due to enzyme denaturation. The most favourable temperature – the point where the enzyme is most active – is known as the **optimum temperature**.

Enzymes are affected by changes in pH. The most favourable pH value – the point where the enzyme is most active – is known as the **optimum pH**. This is not surprising. Amino acid side chains in the active site may act as weak acids and bases with critical functions that depend on their maintaining a certain state of ionization, and elsewhere in the protein ionized side chains may play an essential role in the interactions that maintain protein structure. A less common cause of pH sensitivity is titration of a group on the substrate.

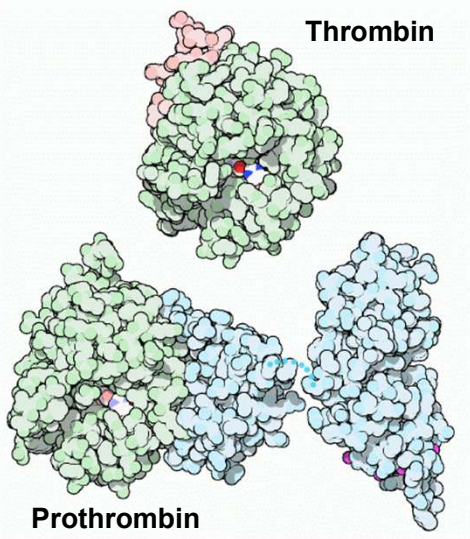
Regulation

- Amount of enzyme
- PTMs
- Inactive precursor
- Allosteric effects
- Cooperativity
- Inhibitors and activators

Cells must regulate the activity of enzymes to coordinate their metabolic processes. Various approaches exist:

- Cells can vary the **amount of enzyme** available via synthesis of new enzyme or degradation of existing enzyme.
- Enzyme activity can be regulated by reversible **PTMs**. For example, serine, threonine, and tyrosine can be modified using phosphorylation.
- For some enzymes, an **inactive precursor**, called a **zymogen**, is cleaved to form the active enzyme.
- **Allosteric effects** arise when binding of substrate or ligand at a site other than active site affects activity or binding. Allosteric effects are possible in a monomeric enzyme, for instance by a simple non-competitive inhibitor, or by binding of an effector to a distant site, perhaps on another subunit.
- **Cooperativity** is the apparent change in affinity or activity of a protein or an enzyme for its ligand or substrate as the concentration of this molecule changes. Cooperativity often requires that the enzyme or protein is built up of interacting subunits. With cooperativity, the substrate itself can be considered an allosteric activator: when it binds to one active site, the activity of the other active sites goes up. The substrate affects active sites far from its binding site.
- Competitive and uncompetitive inhibitors can influence enzymatic activity without allosteric effects.

Example: Zymogens in the Coagulation Cascade



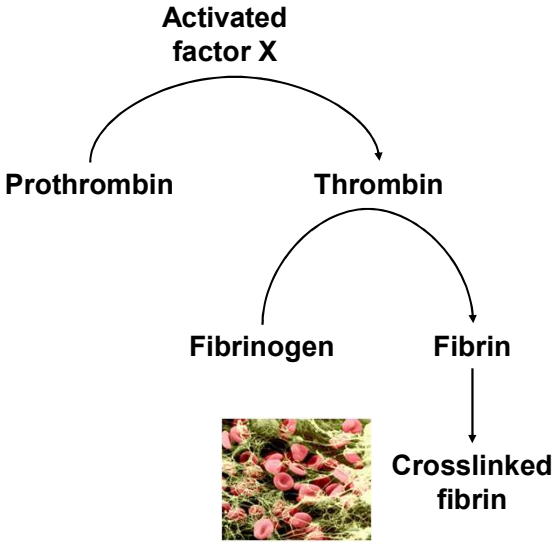


Figure: PDB-101: Molecule of the Month: Thrombin and https://www.britannica.com/science/fibrin

The **coagulation cascade** is a highly regulated stepwise process that ultimately leads to blood clot formation. Each enzyme in the process is initially produced in an inactive form, known as a **zymogen**, and becomes active through proteolytic cleavage. This chain of activations ensures clot formation occurs only when necessary, such as in response to injury.

The final steps of the process start with the activation of **factor X**, which leads to the cleavage of **prothrombin** to **thrombin**. Thrombin then catalyzes the hydrolysis of four Arg-Gly bonds in **fibrinogen** to form **fibrin** monomers that, upon polymerization, form a fibrin clot and stop bleeding.

Prothrombin contains a number of γ-carboxyglutamate residues in its amino-terminal region, introduced by an enzyme that requires **vitamin K**. These carboxyl groups bind Ca^{2+}, which is required to initiate the clotting mechanism. Without sufficient vitamin K, undercarboxylated prothrombin is produced, which lacks the ability to bind Ca^{2+} and is therefore not functional for blood clotting.

Warfarin is a synthetic compound that inhibits the formation of active prothrombin. It inhibits vitamin K oxidoreductase, which generates vitamin K hydroquinone (KH_2) required for vitamin K–dependent proteins. It is particularly poisonous to rats, causing death by internal bleeding. It is however used as an anticoagulant drug for treating humans at risk for excessive blood clotting, such as surgical patients.

Example: Feedback Inhibition

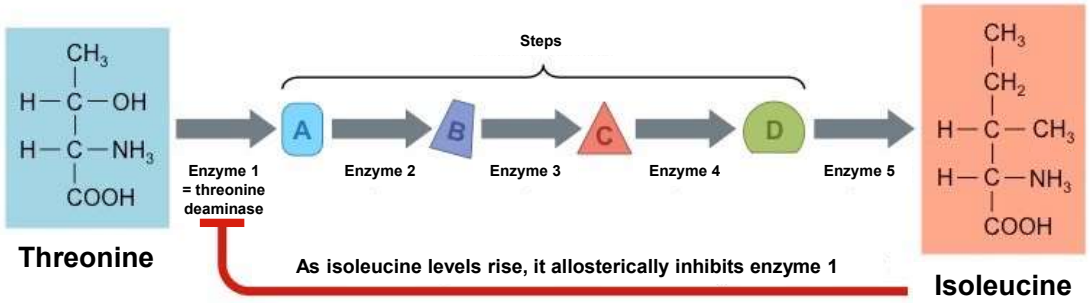


Figure: Feedback Inhibition | BioNinja

In multi-enzyme pathways, the activity of the entire system is regulated by **feedback inhibition**. This means that the final product of the pathway can bind to and inhibit the **regulatory enzyme** (often the first enzyme in the pathway) once its concentration exceeds the cell's needs. By doing so, the cell prevents unnecessary production and conserves resources.

A well-known example of feedback inhibition is the regulation of **isoleucine biosynthesis** in bacteria. In this pathway, the amino acid **isoleucine** is synthesized from **threonine** through a series of enzyme-catalyzed steps. The first enzyme in the pathway, **threonine deaminase**, converts threonine into α-ketobutyrate. When isoleucine levels in the cell become too high, it binds to threonine deaminase and inhibits its activity (allosteric inhibition). This prevents the pathway from continuing and stops further production of isoleucine.

You Need to Be Able to...

- ❑ define an enzyme and its active site
- ❑ compare and identify different types of cofactors
- ❑ classify enzymes according to the types of reaction catalysed
- ❑ explain enzyme energetics and the transition state theory
- ❑ interpret reaction coordinate diagrams
- ❑ understand and apply the relationship between equilibrium constant and (1) the concentrations of each species in the reaction, (2) forward and reverse rate constants, and (3) the free energy change
- ❑ understand and apply the relationship between rate constant and (1) activation energy and (2) reaction rate
- ❑ discuss the origin of an enzyme's catalytic power
- ❑ identify different types of catalytic mechanisms
- ❑ list different substrate-binding models
- ❑ summarize how enzymes are regulated
- ❑ recall factors influencing enzyme activity
- ❑ analyse experimental data to find the optimum temperature and pH for an enzyme

More Information?

- Lehninger Principles of Biochemistry; David L. Nelson, Michael M. Cox
  - 1.3 Physical Foundations
  - 6.1 An Introduction to Enzymes
  - 6.2 How Enzymes Work
- Biochemistry; Jeremy M. Berg, Gregory J. Gatto Jr., Justin K. Jines, John L. Tymoczko, Lubert Stryer
  - 5.1 Enzymes Are Powerful and Highly Specific Catalysts
  - 5.2 Gibbs Free Energy Is a Useful Thermodynamic Function for Understanding Enzymes
  - 5.3 Enzymes Accelerate Reactions by Facilitating the Formation of the Transition State
  - 6.1 Enzymes Use a Core Set of Catalytic Strategies

BIOC0001 – Proteins and Enzymes

Enzyme Kinetics

Dr. Renée Vancraenenbroeck
November 2, 2026

Here, we will investigate how to study the rate of a reaction and determine how it changes in response to experimental parameters (*i.e.* **enzyme kinetics**). This is the oldest yet still most common approach to study the mechanism of an enzyme-catalysed reaction.

Studying enzyme kinetics is useful for

- measuring the concentration of an enzyme in a mixture,
- investigating an enzyme's purity,
- establishing an enzyme's efficiency for different substrates,
- comparing different forms of the same enzyme in different tissues or organisms, and
- evaluating the effects and mechanisms of inhibitors.

Interdisciplinary Approach

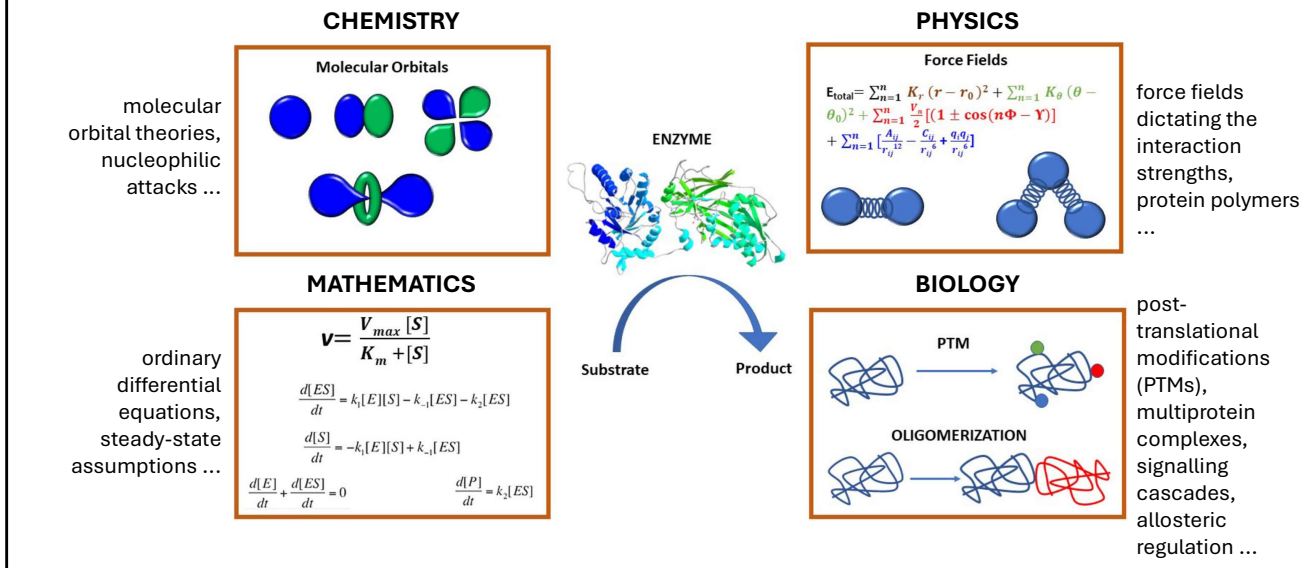


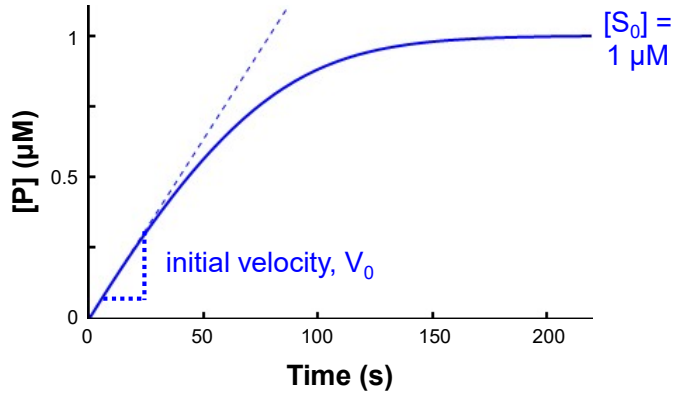
Figure: Words of Advice: teaching enzyme kinetics - Srinivasan - 2021 - The FEBS Journal - Wiley Online Library

Enzymology is **interdisciplinary** and requires knowledge from multiple scientific disciplines. For example:

- Chemistry helps explain catalytic mechanisms
- Physics is essential for predicting protein structure and dynamics using computational tools such as force fields
- Mathematics provides a quantitative framework for analysing enzyme kinetics
- An enzyme operates within a biological context

We will now use some mathematical equations. Don't worry, you don't need to be a maths expert! Just focus on the **concepts** behind the equations, and you'll see how maths helps us make sense of enzymes.

Progress Curve and Initial Velocity, V_0



When following the conversion of substrate S to product P in an enzyme-catalysed reaction one can follow **product concentration ([P]) as a function of time** for one particular initial substrate concentration ($[S_0]$). This type of curve is called a **progress curve**.

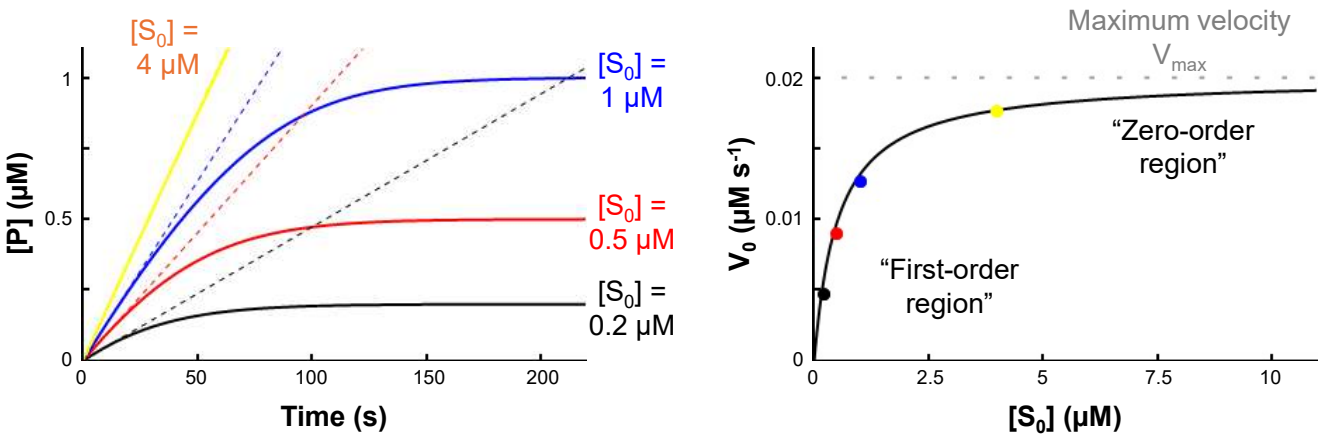
The **reaction rate or velocity** (*i.e.* the slope of the curve = $\Delta[P]/\Delta t$) decreases in time. Some causes for the drop include:

- Substrate is depleted.
- The reverse reaction (*i.e.* P → S) becomes more predominant as product accumulates.
- The products of the reaction inhibit the enzyme.
- The enzyme becomes unstable during the reaction.

If we determine the **initial velocity (V_0)** from the slope of the curve at the beginning of a reaction (often the first 60 seconds or less), we circumvent these issues:

- Changes in the substrate concentration can be limited to a few percent, and the substrate concentration can be regarded as constant, almost equal to the initial substrate concentration: $[S] \approx [S_o]$.
- Very little product has accumulated: $[P] \approx 0$. The reverse reaction can be considered negligible and possible inhibitory effects of product on enzyme activity not significant.

Relationship between [Substrate$_0$] and V_0



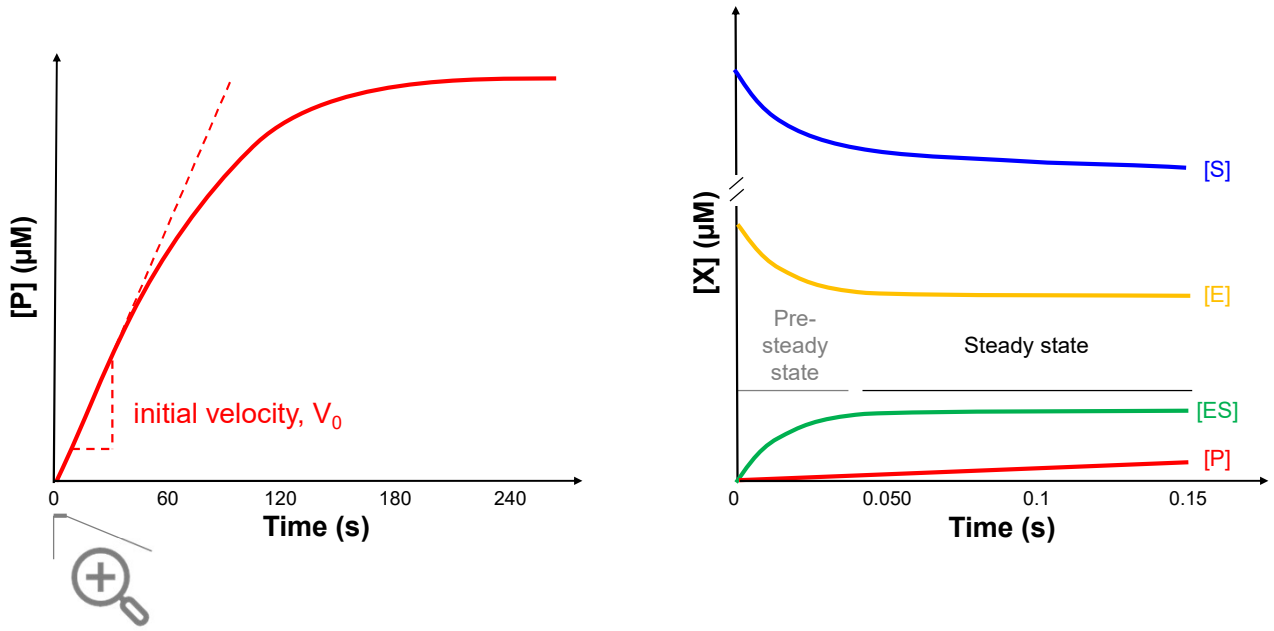
In a typical experiment, we obtain initial velocities for various substrate concentrations. When we then plot the initial velocities as a function of the initial substrate concentrations, we see a **hyperbolic dependence**.

At a fixed concentration of enzyme, the initial velocity is almost linearly proportional to the initial substrate concentration when the initial substrate concentration is small. This part is called the **first-order region** of the curve.
> Remember, first-order kinetics occurs when a constant proportion of A is converted per unit time, -d[A]/dt = k[A]. It is a concentration-dependent process: the higher the concentration of A, the faster the conversion.

However, the initial velocity is nearly independent of the initial substrate concentration when the initial substrate concentration is large. This part is called the **zero-order region** of the curve. This plateau-like initial velocity region is close to the **maximum velocity, V_{max}** (never reached!!).
> Remember, zero-order kinetics occurs when a constant amount of A is converted per unit time, -d[A]/dt = k . Zero-order kinetics is independent in the concentration of A.

Of note, during these experiments the initial enzyme concentration ([E_0]) is much smaller than the initial substrate concentration and is held constant!

Pre-steady State and Steady State



Let us zoom in on the first part of the progress curve. For the reaction E + S ↔ ES → E + P, the enzyme first combines with its substrate to form an enzyme-substrate complex; the enzyme-substrate complex then breaks down in a second step to yield the free enzyme and the reaction product. The enzyme is free to catalyse reaction of another molecule of substrate.

When the enzyme is first mixed with a large excess of substrate, there is an initial period, the **pre–steady state**, during which the concentration of the enzyme-substrate complex builds up. This period is usually too short to be easily observed, lasting just micro- to milliseconds. It can only be measured when we use specialized equipment, such as a stopped-flow instrument.

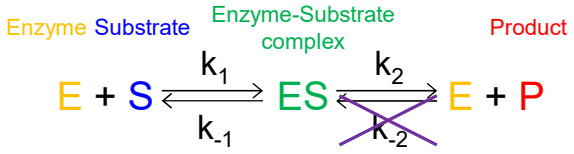
The reaction quickly achieves a **steady state** in which the concentration of enzyme-substrate complex remains approximately constant over time: the enzyme-substrate complex breaks down as quickly as it is formed.

When we determine the slope of the progress using the first two minutes or less, we measure the initial velocity reflecting the steady state. Analysis of these initial velocities is referred to as **steady state kinetics.**

Assumptions

Assumption 3:
The initial velocity is determined by [ES]: $V_o = k_2$ [ES]

Assumption 2 – steady-state assumption:
At steady state, [ES] is constant.



Assumption 1:
The conversion of P to S is negligible.

Assumption 4:
Changes in [S] are very small: [S] ≈ $[S_o]$

Assumption 5:
The enzyme only exist in two forms: $[E_o]$ = [E] + [ES]

We now develop the basic logic and the algebraic steps in a modern derivation of the Michaelis-Menten equation or rate equation for a one-substrate catalysed reaction.

We use the following **assumptions**:
1. At the beginning of the reaction, very little product has accumulated. The reverse reaction, *i.e.* the conversion of product back to substrate, can be considered negligible. The reverse rate constant, k_{-2}, is ignored.
2. At steady state, the concentration of enzyme-substrate complex is constant: the enzyme-substrate complex breaks down as quickly as it is formed. This assumption is called the **steady-state assumption** and was introduced by **Briggs and Haldane**.
3. The binding step (E + S ↔ ES) is fast, allowing the reaction to quickly reach equilibrium ratios of [E], [S], and [ES]. The catalytic step (ES → E + P) is slower, and thus rate-limiting. Therefore, the overall rate must be proportional to the concentration of the species that reacts in the second step, that is the enzyme-substrate complex.
4. The initial enzyme concentration, $[E_0]$, is much smaller than the initial substrate concentration, $[S_0]$. Changes in the substrate concentration can be limited to a few percent and the substrate concentration can be regarded as constant: [S] ≈ $[S_o]$.
5. The enzyme exists in only two forms: free enzyme, E, and substrate-bound enzyme, ES. Thus, the total enzyme concentration, *i.e.* the initial enzyme concentration $[E_0]$, is the sum of the free and substrate-bound concentrations: $[E_0]$ = [E] + [ES].

Derivation of Michaelis-Menten Equation

Rate of formation of ES $= k_1\,[E][S]$ *eq 1*

Rate of breakdown of ES $= (k_{-1} + k_2)[ES]$ *eq 2*

Steady state ↓

Rate of formation of ES = Rate of breakdown of ES *eq 3*

$k_1[E][S] = (k_{-1} + k_2)[ES]$ *eq 4*

$[E_0] = [E] + [ES]$; $[S_0] >>> [E_0]$, $[S] \approx [S_0]$

$k_1([E_0] - [ES])[S_0] = (k_{-1} + k_2)[ES]$ *eq 5*

Rearrange ↓

$$[ES] = \frac{k_1[E_0][S_0]}{k_1[S_0] + k_{-1} + k_2}$$ *eq 6*

Simplify ↓

$$[ES] = \frac{[E_0][S_0]}{[S_0] + \left(\frac{k_{-1} + k_2}{k_1}\right)}$$ *eq 7*

$\frac{k_{-1} + k_2}{k_1} = K_m$ ↓

$$[ES] = \frac{[E_0][S_0]}{[S_0] + K_m}$$ *eq 8*

$V_0 = k_2[ES]$ *eq 9*

Combine ↓

$$V_0 = \frac{k_2[E_0][S_0]}{[S_0] + K_m}$$ *eq 10*

$V_{max} = k_2[E_0]$ ↓

$$V_0 = \frac{V_{max}[S_0]}{[S_0] + K_m}$$ *eq 11*

Using the assumptions, we can derive the Michaelis-Menten equation:

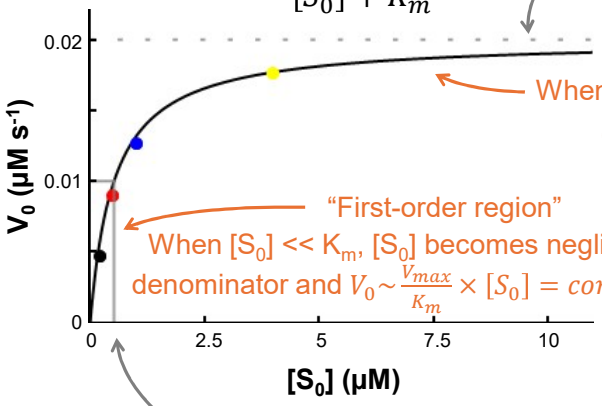
- The rates of formation and breakdown of the enzyme-substrate complex are determined by the steps described by the rate constants k_1 (formation) and k_{-1} and k_2 (breakdown), according to *(eq 1)* and *(eq 2)*.
- We now use the steady-state assumption: the concentration of the enzyme-substrate complex is constant - that is, the rate of formation of enzyme-substrate complex is equal to the rate of its breakdown. This leads to *(eq 3)* and *(eq 4)*.
- Since substrate is present in large excess over enzyme, we assume that the free substrate concentration is well approximated by the total (*i.e.* initial) substrate concentration added to the reaction, $[S_0]$. We also assume that the enzyme exists in only two forms: free enzyme, E, and substrate-bound enzyme, ES. The free enzyme concentration, [E], is given by the difference between the total (*i.e.* initial) enzyme concentration, $[E_0]$, and the free concentration of the enzyme-substrate complex, [ES]. This leads to *(eq 5)*.
- We now solve *(eq 5)* for the concentration of the enzyme-substrate complex. This *(eq 6)* is simplified further, combining all rate constants in one expression.
- The term with rate constants in *(eq 7)* is substituted by the **Michaelis constant, K_m** *(Michaelis, Menten, Briggs and Haldane introduced this constant!)* giving *(eq 8)*.
- We assume a fast binding step and a slow catalytic step. The overall rate must be proportional to the enzyme-substrate complex *(eq 9)*. We can now substitute the concentration of the enzyme-substrate complex by *(eq 8)*.
- Combining the two equations *(eq 8 and eq 9)*, we obtain the following relation *(eq 10)*. This describes the initial reaction velocity as a hyperbolic function of initial substrate

concentration, $[S_0]$.

- At infinite initial substrate concentration ($[S_0] \rightarrow \infty$), the enzyme is saturated ($[ES] \approx [E_0]$) and the initial velocity approaches the **maximum velocity, $V_{max} = k_2 [E_0]$**.
- We now have the **Michaelis-Menten equation** in *(eq 11)*.

Understanding the Michaelis-Menten Equation

$$V_0 = \frac{V_{max}[S_0]}{[S_0] + K_m}$$



V_{max} (maximum velocity), here 0.02 μM s^{-1}
The initial reaction velocity when $[S_0] >> K_m$

"Zero-order region"
When $[S_0] >> K_m$, K_m becomes negligible in the denominator and $V_0 \sim \frac{V_{max}[S_0]}{[S_0]} = V_{max}$

"First-order region"
When $[S_0] << K_m$, $[S_0]$ becomes negligible in the denominator and $V_0 \sim \frac{V_{max}}{K_m} \times [S_0] = constant \times [S_0]$

K_m (Michaelis-Menten constant), here 0.5 μM
The substrate concentration resulting in half-maximal velocity *(i.e.* $\frac{V_{max}}{2} = \frac{V_{max}[S_0]}{[S_0]+K_m} \rightarrow K_m = [S_0]$*)*

Using the **Michaelis-Menten equation**, we can explain the observed behavior.

At any given instant in an enzyme-catalysed reaction, the enzyme exists in two forms that are in equilibrium, the free form, E, and the substrate-bound form, ES.

At low initial substrate concentration, most of the enzyme is in the free form. Here, in the **first-order region** *(when $[S_0] << K_m$, $V_0 \sim (V_{max}/K_m) \times [S_0]$)*, the rate is almost linearly proportional – with V_{max}/K_m the proportionality constant – to the initial substrate concentration: the equilibrium is pushed toward formation of more enzyme-substrate complex as the initial substrate concentration increases.

When the initial substrate concentration is sufficiently high that essentially all the free enzyme has been converted to the enzyme-substrate form, we are close to the maximum initial velocity (V_{max}) *(when $[S_0] >> K_m$, $V_0 \sim V_{max}$)*. Under these conditions, called the **zero-order region** of the curve, **the enzyme is "saturated" with its substrate**, so that further increases in the initial substrate concentration have no effect on rate. The saturation effect is a distinguishing characteristic of enzymatic catalysts!

To summarize, the initial phase of the reaction where the free form of enzyme, E, dominates is controlled by the rate constant V_{max}/K_m (representing E + S ↔ ES). The latter phase, where the substrate-bound form, ES, dominates, is related to V_{max} (representing ES → E + P).

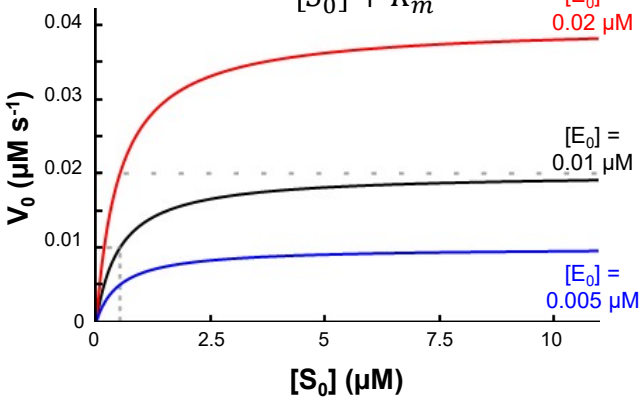
Let us now consider the situation when the initial substrate concentration being evaluated

results in an initial velocity that is exactly ½ of the maximum velocity ($V_0 = V_{max} / 2$). We reorganize the equation and get $[S_0] = K_m$. Thus, **the Michaelis constant K_m is the initial substrate concentration that results in ½ of the maximum velocity**.

Understanding the Michaelis-Menten Equation

(cont.)

$$V_0 = \frac{V_{max}[S_0]}{[S_0] + K_m}$$



As $V_{max} = k_2[E_0]$, the initial velocity at any single substrate concentration is proportional to $[E_0]$

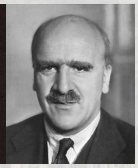
As $V_{max} = k_2 [E_0]$, **the initial velocity is directly proportional to the initial enzyme concentration**: if we double the total initial enzyme concentration, we double the initial velocity at any single initial substrate concentration, $[S_0]$.

Michaelis-Menten Equation and Parameters

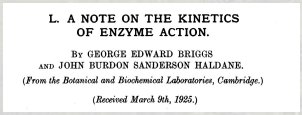

L Michaelis (1875-1949)


M Menten (1879-1960)


GE Briggs (1893-1985)


JBS Haldane (1892-1964)


Die Kinetik der Invertinwirkung
Von
L. Michaelis and Miss Maud L. Menten
(Received 4 February 1913.)
With 19 Figures in Text.
The Kinetics of Invertase Action
translated by
Roger S. Goody[1] and Kenneth A. Johnson[2]


L. A NOTE ON THE KINETICS OF ENZYME ACTION.
By GEORGE EDWARD BRIGGS and JOHN BURDON SANDERSON HALDANE.
(From the Botanical and Biochemical Laboratories, Cambridge.)
(Received March 9th, 1925.)

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

$$V_0 = \frac{V_{max}[S_0]}{[S_0] + K_m}$$

$$V_{max} = k_{cat}[E_0]$$

here, k_2= turnover number, k_{cat} (substrate molecules converted per enzyme per unit time at saturation)

$$K_m = \frac{k_{-1} + k_2}{k_1}$$

$$K_d = \frac{k_{-1}}{k_1}$$

defines the relationship between the initial velocities and the initial substrate concentrations

V_{max} and k_{cat} provide information on substrate turnover

K_m provides information on substrate binding

Mathematical basis has been developed before the insights into 3D structures of proteins, including enzymes!

Meaning of K_m

Graphically, the Michaelis constant K_m *(units: M)* is the initial substrate concentration that results in ½ of the maximum velocity.

K_m is a property associated with **binding of substrate to enzyme** but *it cannot be considered a simple measure of substrate affinity as* $K_m = (k_{-1}+k_2)/k_1$! The dissociation constant of the enzyme-substrate complex, K_d, is k_{-1}/k_1. We can see that, when k_2 is small compared to k_{-1}, K_m is a decent approximation of the K_d.

A well-optimized enzyme has a **$[S_0]/K_m$ ratio typically around 1.0 under physiological conditions**. If K_m <<< $[S_0]$, the enzyme would be saturated under all conditions and could not detect / convert an increase in $[S_0]$. If K_m >>> $[S_0]$, most of the enzyme is in its free state, which is a waste of resources.

Meaning of V_{max} and k_{cat}

Graphically, the maximal velocity V_{max} *(units: M s^{-1})* is the initial velocity when the enzyme is "saturated" with its substrate, so at high initial substrate concentrations.

V_{max} is a property associated with **the enzyme's ability to turn over substrate**.

From V_{max}, we can calculate the **turnover number $k_{cat} = V_{max} / [E_0]$ *(units: s^{-1})*** of an enzyme, which is **the number of substrate molecules converted to product in a given unit of time on a single enzyme molecule when the enzyme is saturated with substrate**.

In the classic Michaelis-Menten mechanism, $k_{cat} = k_2$. Earlier, we used k_{cat} to calculate the rate enhancement (k_{cat} / k_{uncat}).

Specificity Constant

Enzyme	Substrate	*Michaelis constant* K_M (M)	*Turnover number* k_{cat} (s^{-1})	*Specificity constant* k_{cat}/K_M ($M^{-1} \cdot s^{-1}$)
Acetylcholinesterase	Acetylcholine	9.5×10^{-5}	1.4×10^{4}	1.5×10^{8}
Carbonic anhydrase	CO_2	1.2×10^{-2}	1.0×10^{6}	8.3×10^{7}
	HCO_3^-	2.6×10^{-2}	4.0×10^{5}	1.5×10^{7}
Catalase	H_2O_2	2.5×10^{-2}	1.0×10^{7}	4.0×10^{8}
Chymotrypsin	*N*-Acetylglycine ethyl ester	4.4×10^{-1}	5.1×10^{-2}	1.2×10^{-1}
	N-Acetylvaline ethyl ester	8.8×10^{-2}	1.7×10^{-1}	1.9
	N-Acetyltyrosine ethyl ester	6.6×10^{-4}	1.9×10^{2}	2.9×10^{5}
Fumarase	Fumarate	5.0×10^{-6}	8.0×10^{2}	1.6×10^{8}
	Malate	2.5×10^{-5}	9.0×10^{2}	3.6×10^{7}
Urease	Urea	2.5×10^{-2}	1.0×10^{4}	4.0×10^{5}

Upper limit *(~ 10^8 – 10^9 M^{-1} s^{-1}), the rate at which S and E* ***diffuse*** *together*

Preferred substrate

Table: Voet & Voet, Biochemistry, John Wiley and Sons

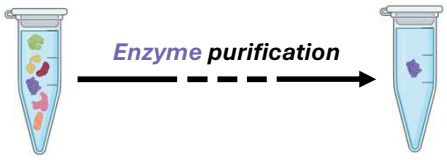
The **specificity constant or enzyme efficiency, k_{cat}/K_m** describes **how efficient the enzyme is at converting free substrate to free product**. Neither k_{cat} nor K_m are suitable individually to compare catalytic function since the former neglects the substrate binding and the latter neglects the catalysis.

There is a **natural upper limit** to this efficiency: it can't be faster than the rate at which the enzyme and substrate encounter each other in solution. This is called the diffusion limit. The fastest enzymes have k_{cat}/K_m values close to **10^8 to 10^9 M^{-1} s^{-1}**. These enzymes are so efficient that they catalyse a reaction almost every time they meet a substrate molecule.

When an enzyme can act on **more than one substrate**, we use the ratio **k_{cat}/K_m** to compare how efficiently it works with each one. The higher the specificity constant, the more the enzyme "likes" or **prefers** that substrate.

Specific Activity

$$Specific\ activity\left(^{U}/_{mg}\right) = \frac{Total\ activity\ \left(^{U}/_{ml}\right)}{Total\ protein\left(^{mg}/_{ml}\right)} \quad \text{with} \quad U = {}^{\mu mol}/_{min}$$



* **Total protein (mg)** – *how much protein present in the sample* = [Protein] (mg/ml) as determined using protein concentration assay x Volume (ml)

* **Total activity (U)** – *how much µmol product formed per minute in the samplw* = Activity (U/ml) as determined using enzyme activity assay x Volume (ml)

* **Specific activity (U/mg)** – *how pure the sample is* = Total activity (U) / Total protein (mg)

* **Yield (%)** – *the enzyme activity retained after each step* = Total activity (step *n*) / Total activity (first step) x 100 %

Step	Total protein (mg)	Total activity (U)	Specific activity (U/mg)	Yield (%)
Crude lysate	1250	118	0.09	100
Crude extract	967	106	0.11	90
Ammonium sulphate precipitation	180	75	0.42	64
DEAE column (IEX)	24	60	2.50	51
Sephacryl column (GF)	3.6	46	12.8	39

$$64\ \% = {}^{75}/_{118} \times 100\ \%$$

Specific activity tells us how active an enzyme sample is **relative to how much protein** is present. It is calculated as the total activity of a sample (expressed in units per ml) divided by the total concentration of protein (expressed in mg/ml). Specific activity values are therefore quoted as **units/mg**.

One unit (U) is the amount if enzyme that catalyses the reaction of 1 µmol (some labs use 1 nmol) of substrate per minute under standard conditions. Unfortunately, the term 'standard conditions' is open to interpretation.

Specific activity is **an important measure of enzyme purity**. After each purification step, the total protein concentration and the activity of the sample is measured, the ratio of the two gives the specific activity. Activity and total protein generally decrease with each step. In a successful step, the loss of nonspecific protein is much greater than the loss of activity; therefore, specific activity increases even as total activity falls.

When an enzyme is purified as described before and one has a specific activity that is below the expected specific activity value based on previous records, the sample may contain impurities or enzyme molecules that have become denatured.

Lineweaver-Burk Plot

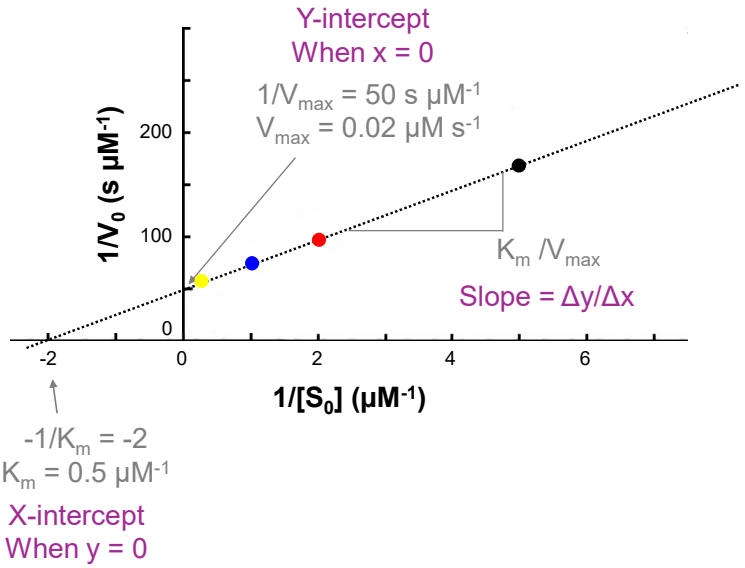
$$V_0 = \frac{V_{max}[S_0]}{[S_0] + K_m}$$

Reciprocal ↓

$$\frac{1}{V_0} = \frac{[S_0] + K_m}{V_{max}[S_0]}$$

Rearrange ↓

$$\frac{1}{V_0} = \frac{K_m}{V_{max}[S_0]} + \frac{1}{V_{max}}$$



Before the availability of computers, the determination of K_m and V_{max} values required algebraic manipulation of the basic Michaelis-Menten equation.

One can transform the Michaelis-Menten equation into an equation that gives a straight-line plot by taking the reciprocal of both sides of the equation. This equation gives a plot of $1/V_0$ versus $1/[S_0]$, called a **Lineweaver-Burk or double-reciprocal plot**. We get a straight line with a *Y*-intercept of $1/V_{max}$ and a slope of K_m/V_{max}. The *X*-intercept is $-1/K_m$.

As a double reciprocal plot, the Lineweaver-Burk plot presents two problems when used with real life experimental data. First, all data found at large substrate concentrations will be clustered near the origin. Second, small experimental errors are magnified on Lineweaver-Burk plots, particularly for the points that lie far away from the origin. Thus, relying too heavily on the points far from the origin can lead to inaccurate values of K_m and V_{max}.

You Need to Be Able to…

- ❑ elaborate on how enzyme kinetics can be used
- ❑ explain the meaning of enzyme initial velocities and how they are determined
- ❑ analyse progress curves
- ❑ describe the relationship between initial velocity and substrate concentration
- ❑ proof and employ the Michaelis-Menten equation
- ❑ calculate and discriminate between K_m, V_{max}, k_{cat}, specificity constant, rate enhancement, and specific activity
- ❑ construct and interpret Michaelis-Menten and Lineweaver-Burke plots

More Information?

- Lehninger Principles of Biochemistry; David L. Nelson, Michael M. Cox
  - 6.3 Enzyme Kinetics as an Approach to Understanding Mechanism
- Biochemistry; Jeremy M. Berg, Gregory J. Gatto Jr., Justin K. Jines, John L. Tymoczko, Lubert Stryer
  - 5.4 The Michaelis-Menten Model Accounts for the Kinetic Properties of Many Enzymes

BIOC0001 – Proteins and Enzymes

Enzyme Inhibitors

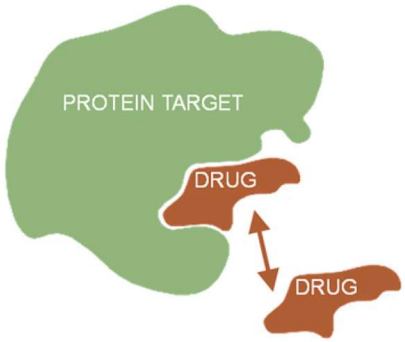
Dr. Renée Vancraenenbroeck
November 2, 2026



Here, we will investigate how enzyme kinetics can be used to evaluate the effects and mechanisms of **inhibitors**.

Reversible and Irreversible, Covalent Inhibitors

Reversible Inhibitor (Drug)
In equilibrium with protein target



Irreversible, Covalent Inhibitor (Drug)
Permanently bound to protein target

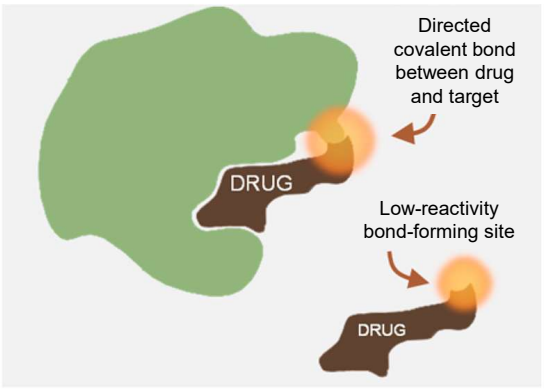


Figure: The Ascension of Targeted Covalent Inhibitors | Journal of Medicinal Chemistry

Enzyme inhibitors are molecules interfering with enzyme catalysis. They can be **natural products or synthetic compounds** – many **drugs** are enzyme inhibitors!

There are two broad classes of enzyme inhibitors: irreversible and reversible.

- An **irreversible** inhibitor *dissociates very slowly* – if at all – from its target enzyme because it has become tightly bound to the enzyme, mostly through a covalent bond. Each inhibition event effectively removes a molecule of enzyme from availability. **Covalent inhibition** is a famous example.
- **Reversible** inhibition on the other hand, is characterized by a *rapid dissociation* of the enzyme-inhibitor complex. There are four major subdivisions of reversible inhibition: **competitive, uncompetitive, noncompetitive, and mixed inhibition**.

All these inhibitors interact differently with the enzyme or enzyme-substrate complex, and they have different effects on enzyme activity. We can tell which type of inhibition is occurring by looking how the values of V_{max}, V_{max}/K_m, and K_m change in the presence of inhibitor!

Direct Plot and Lineweaver-Burk Plot

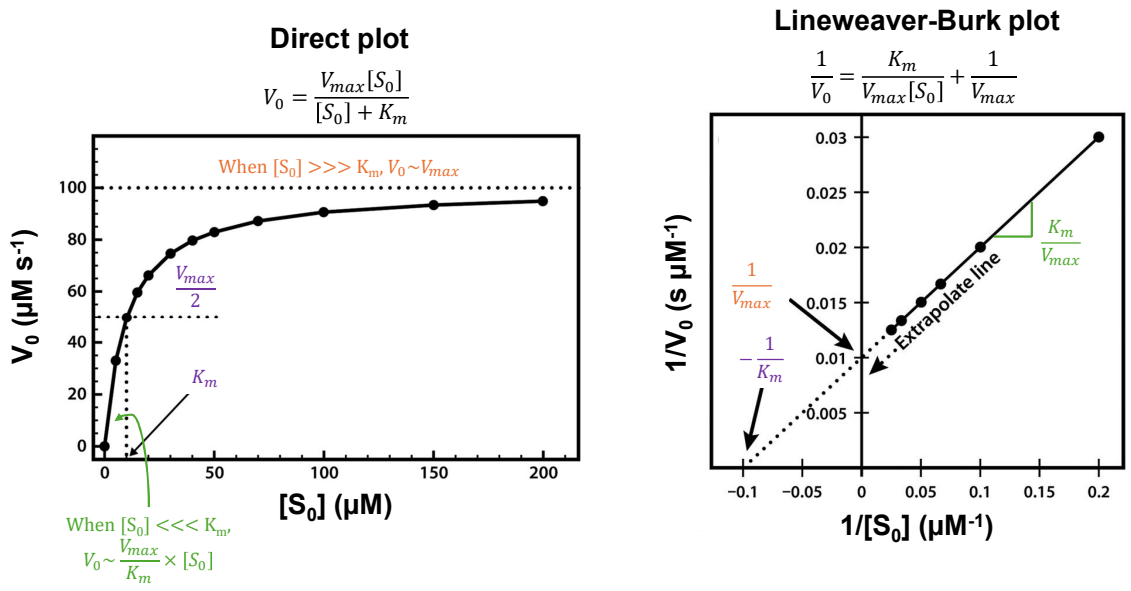


Figure: Enzymes: principles and biotechnological applications | Essays in Biochemistry | Portland Press

Remember that:

- V_{max}, the maximum velocity, is the rate at infinitely high substrate concentrations (when $[S_0] >> K_m$, $V_0 \sim V_{max}$). V_{max} is a property associated with the enzyme's ability to turn over substrate.
- K_m, the Michaelis-Menten constant, is the initial substrate concentration that results in ½ of the maximum velocity (at $V_0 = V_{max}/2$, $[S_0] = K_m$). K_m is a property associated with binding of substrate to enzyme.
- V_{max}/K_m is the rate constant proportional to the initial first-order phase (when $[S_0] << K_m$, $V_0 \sim V_{max}/K_m$). Please note that V_{max}/K_m is the reciprocal of the slope in the Lineweaver-Burk plot.

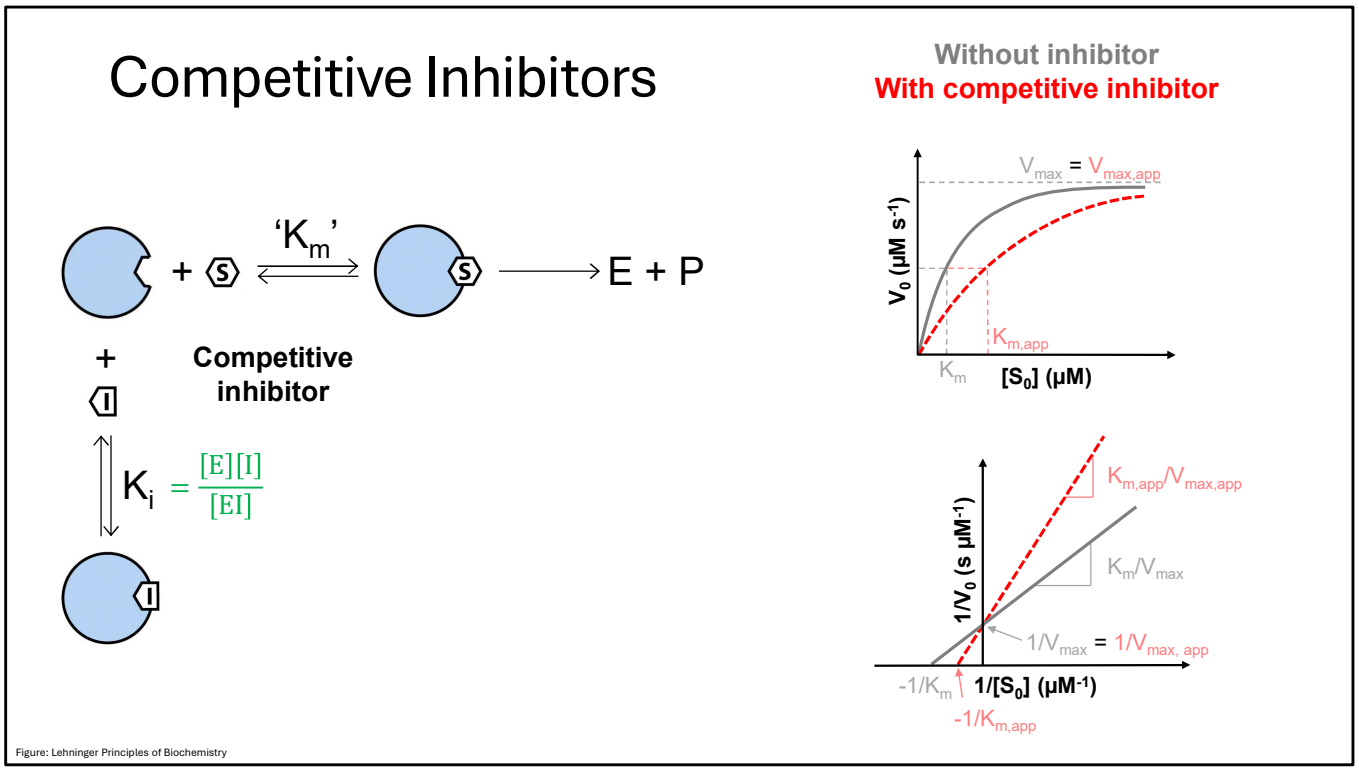


Figure: Lehninger Principles of Biochemistry

A **competitive inhibitor** competes with substrate for binding to the active site. A competitive inhibitor often resembles the substrate in structure.

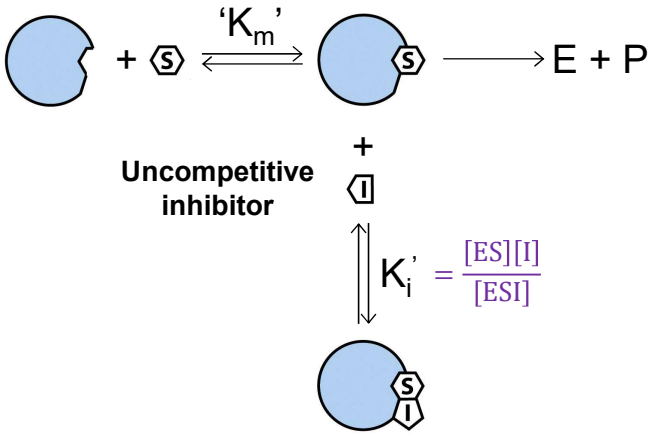
In the presence of competitive inhibitor, an enzyme-inhibitor complex is formed, thereby reducing free enzyme that is available for substrate binding. A competitive inhibitor hinders binding of the substrate and increases the Michaelis constant (K_m). Once enzyme-substrate complex is formed, the turnover of substrate is unaffected. A competitive inhibitor does not change the maximum velocity (V_{max}).

Competitive inhibitors do not affect the V_{max} asymptote of the curve at $[S_0] >> K_m$ of the direct plot. They do decrease the V_{max}/K_m tangent to the curve at $[S_0] << K_m$. Competitive inhibitors do not alter the y-intercept ($1/V_{max}$) of a Lineweaver-Burk plot. They do increase the slope (K_m/V_{max}).

A competitive inhibitor only has an effect at very low substrate concentrations. The effects of competitive inhibition can be overcome by increasing substrate concentration, which naturally happens when we inhibit a metabolic process. Hence, a system will be able to adjust with little difficulty to competitive inhibition.

Of note, the enzyme in the absence of inhibitor is characterised by V_{max}, V_{max}/K_m, and K_m; the enzyme in the presence of inhibitor is characterised by $V_{max,app}$, $V_{max,app}/K_{m,app}$, and $K_{m,app}$. The "$_{app}$" stands for "apparent", indicating that the measured value changes because of the inhibitor.

Uncompetitive Inhibitors



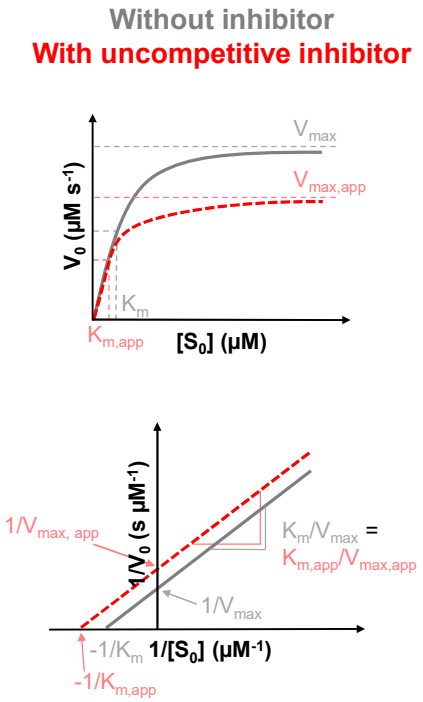


Figure: Lehninger Principles of Biochemistry

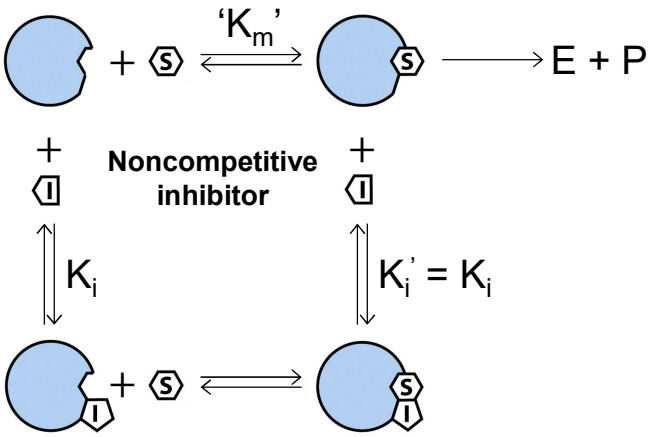
An **uncompetitive inhibitor** binds to the enzyme-substrate complex, at a location other than the active site. The inhibitor can bind on a different site on the enzyme, only accessible after substrate binding; or the inhibitor can bind to groups on both the enzyme and the substrate.

In the presence of uncompetitive inhibitor, the inhibitor will draw off some of the enzyme-substrate complex from the pathway, thereby driving the E+S → ES binding reaction. An uncompetitive inhibitor decreases the Michaelis constant (K_m). After uncompetitive inhibitor binding to the enzyme-substrate complex, we have an unproductive enzyme-substrate-inhibitor complex and substrate turnover is affected. An uncompetitive inhibitor decreases the maximum velocity (V_{max}).

Uncompetitive inhibitors lower the V_{max} asymptote of the curve at $[S_0] >> K_m$ of the direct plot. They do not decrease the V_{max}/K_m tangent to the curve at $[S_0] << K_m$. Uncompetitive inhibitors increase the y-intercept ($1/V_{max}$) of a Lineweaver-Burk plot. They do not change the slope (K_m/V_{max}).

An uncompetitive inhibitor only has an effect at high substrate concentrations. The effects of uncompetitive inhibition cannot be overcome by increasing substrate concentration, which naturally happens when we inhibit a metabolic process. Hence, a system cannot easily adjust to uncompetitive inhibition.

Noncompetitive Inhibitors



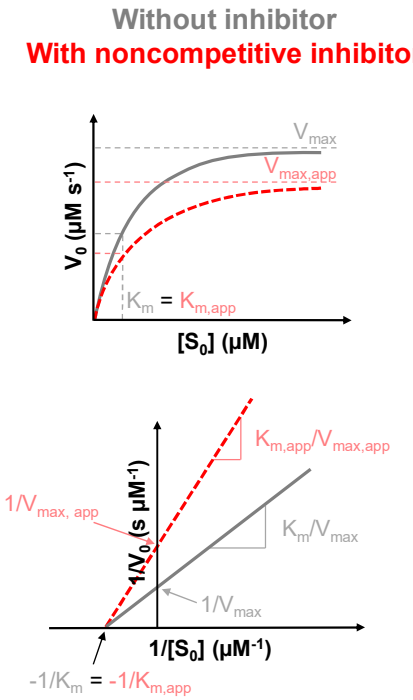


Figure: Lehninger Principles of Biochemistry

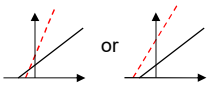
A **noncompetitive inhibitor** binds to the enzyme and enzyme-substrate complex with the same affinity. The enzyme has a permanent site that the inhibitor can bind to, regardless of whether the substrate is bound to the active site.

Adding a noncompetitive inhibitor is like subtracting active enzyme molecules, both free enzyme and enzyme-substrate complex as the inhibitor can bind to both the enzyme and the enzyme-substrate complex with the same affinity. The Michaelis constant (K_m) does not change but the maximum velocity (V_{max}) decreases.

Noncompetitive inhibitors lower the V_{max} asymptote of the curve at $[S_0] >> K_m$ of the direct plot. They decrease the V_{max}/K_m tangent to the curve at $[S_0] << K_m$. Noncompetitive inhibitors increase the y-intercept ($1/V_{max}$) and the slope (K_m/V_{max}) of a Lineweaver-Burk plot.

A noncompetitive inhibitor has an effect at low and high substrate concentrations. The effects of noncompetitive inhibition cannot be overcome by increasing substrate concentration, which naturally happens when we inhibit a metabolic process. Hence, a system cannot easily adjust to noncompetitive inhibition.

Reversible Inhibitors: Summary

Inhibition	Influence	Lineweaver-Burk plot Without inhibitor With inhibitor	In the presence of inhibitor, we get $V_0 = \frac{V_{max,app}[S_0]}{K_{m,app} + [S_0]}$ with $V_{max,app} =$	$K_{m,app} =$
Competitive	V_{max} =; K_m ↗; K_m/V_{max} ↗		V_{max}	$\left(1+\frac{[I]}{K_i}\right) K_m$; α
Uncompetitive	V_{max} ↘ ; K_m ↘; K_m/V_{max} =		$\frac{V_{max}}{\left(1+\frac{[I]}{K_i'}\right)}$; α'	$\frac{K_m}{\left(1+\frac{[I]}{K_i'}\right)}$
Noncompetitive $K_i = K_i'$	V_{max} ↘; K_m =; K_m/V_{max} ↗		$\frac{V_{max}}{\left(1+\frac{[I]}{K_i'}\right)}$	K_m
Mixed $K_i \neq K_i'$	V_{max} ↘; K_m ≠; K_m/V_{max} ↗	 or	$\frac{V_{max}}{\left(1+\frac{[I]}{K_i'}\right)}$	$\frac{\left(1+\frac{[I]}{K_i}\right) K_m}{\left(1+\frac{[I]}{K_i'}\right)}$

We can quantitatively describe the effects of an inhibitor on V_{max}, V_{max}/K_m, and K_m using the **equilibrium constant between free enzyme, free inhibitor, and the enzyme–inhibitor complex (K_i)**, and the **equilibrium constant between enzyme-substrate complex, free inhibitor, and the enzyme-substrate-inhibitor complex (K_i')**. The enzyme in the presence of inhibitor is characterised by $V_{max,app}$, $V_{max,app}/K_{m,app}$, and $K_{m,app}$.

- Competitive inhibition: V_{max} does not change. K_m is multiplied by a **factor α**, which relates to the equilibrium constant between free enzyme, free inhibitor, and the enzyme–inhibitor complex (K_i) and the inhibitor concentration.
- Uncompetitive inhibition: Both V_{max} and K_m are divided by **a factor α'**, which relates to the equilibrium constant between enzyme-substrate complex, free inhibitor, and the enzyme-substrate-inhibitor complex (K_i') and the inhibitor concentration.
- Non-competitive inhibition: V_{max} is divided by a factor α'. K_m does not change.

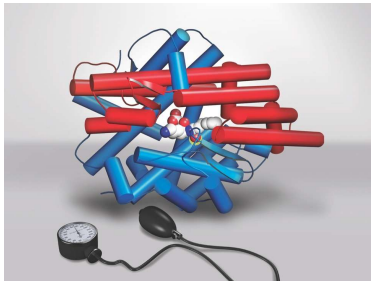
K_i and K_i' can be determined:

- Competitive inhibition: K_i is the concentration of inhibitor that *doubles* K_m.
- Uncompetitive inhibition: K_i' is the concentration of inhibitor that *halves* V_{max} *and* K_m.
- Non-competitive inhibition: $K_i = K_i$' is the concentration of inhibitor that *halves* V_{max}.

Of note, **mixed inhibition** is similar to noncompetitive inhibition except that binding of the substrate or the inhibitor affects the enzyme's binding affinity for the other: the dissociation constants of the inhibitor for the enzyme and the enzyme-substrate complex are different. The Michaelis constant (K_m) changes as the inhibitor can bind to both the enzyme and the enzyme-substrate complex with different affinity. K_m increases when the inhibitor has a

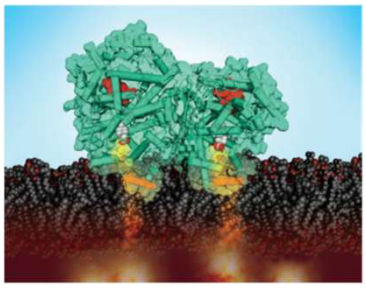
higher affinity for the enzyme *(like competitive inhibition)* and decreases when the inhibitor has a higher affinity for the enzyme-substrate complex *(like uncompetitive inhibition)*. The maximum velocity (V_{max}) always decreases.

Reversible Inhibitors: Examples



Angiotensin Converting Enzyme (homo-dimer, red and blue) + the competitive inhibitor **lisinopril** (marketed as Prinivil; white)

Reduces **high blood pressure** by blocking angiotensin II production



Cyclooxygenase 2 (homo-dimer, green) + the cofactor heme (red) + the competitive inhibitor **ibuprofen** (white)

Reduces **pain and inflammation** by blocking prostaglandin synthesis



Reverse transcriptase (green) + the covalent inhibitor **zidovudine** (pink - incorporated into the nascent DNA strand by the enzyme and thereby blocking further transcription with its 3'-azide modification (blue)) and the noncompetitive inhibitor **nevirapine** (yellow – binding away from the active site)

Reduce the **amount of HIV** in the body by inhibiting reverse transcriptase copying the viral RNA genome and making a DNA copy, necessary for HIV replication

Figures: PDB-101: Learn: Flyers, Posters, & Calendars: Calendars: 2016 A Year in Protein-Drug Complexes

Many drugs are competitive inhibitors:

- Lisinopril is a competitive inhibitor of angiotensin converting enzyme**.** The hormone angiotensin II binds to receptors on blood vessel cells, causing them to constrict and increase blood pressure. Lisinopril is used to lower blood pressure. See https://pdb101.rcsb.org/motm/298.
- The pain killer ibuprofen is a competitive inhibitor of cyclooxygenase (COX) 1 and 2. Both create prostaglandins which are used for basic housekeeping or signalling and inflammation. While inhibiting COX-2 is responsible for its anti-inflammatory and pain-relieving effects, its simultaneous inhibition of COX-1 is associated with gastrointestinal side effects. Therefore, it is a less selective drug than drugs specifically designed to only target COX-2.
- Methotrexate is widely used in the therapy of cancer. It is a competitive inhibitor of dihydrofolate reductase, an enzyme that provides key metabolites for nucleotide biosynthesis. Methotrexate resembles the natural substrate of dihydrofolate reductase, folate. See https://pdb101.rcsb.org/motm/34.
- Allopurinol is widely used for the treatment of gout. It is a competitive inhibitor of xanthine oxidase, an enzyme that produces uric acid by breaking down ATP and GTP. The uric acid itself, as well as the reactive oxygen species released during the enzymatic reaction, can have detrimental effects on the body. Allopurinol resembles the natural substrates of xanthine oxidase, xanthine and hypoxanthine. See https://pdb101.rcsb.org/motm/117.
- Statins are widely used for treating hypercholesterolemia. They are competitive inhibitors of 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase. This

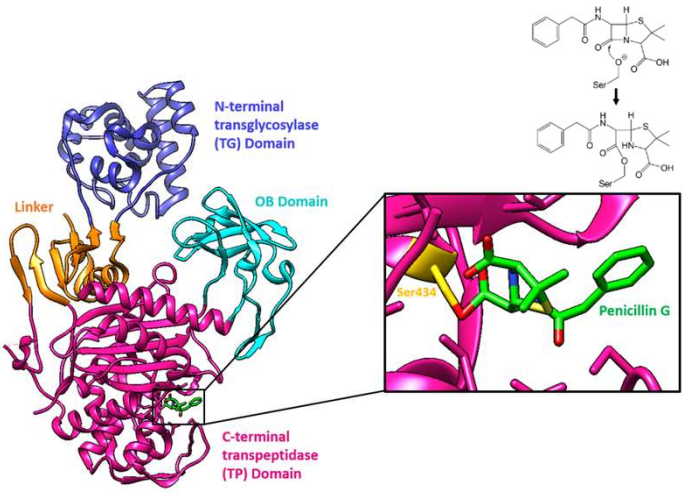
enzyme catalyzes the conversion of HMG-CoA to mevalonate, a rate-limiting step in the synthesis of cholesterol. The statins occupy a portion of the binding site of HMG-CoA, thus blocking access of this substrate to the active site. See https://pdb101.rcsb.org/motm/96.

- Imatinib is widely used in the therapy of cancer. It is a competitive inhibitor of Bcr-Abl tyrosine kinase, an enzyme that drives cell division. Imatinib occupies the same binding site as ATP, thus blocking access of this substrate to the active site. See https://pdb101.rcsb.org/motm/283.

Uncompetitive inhibitor drugs are very rare. One example is the use of lithium to ameliorate manic-depressive psychosis. The specific mechanisms by which lithium exerts its mood-stabilizing effects are not well understood. It is known that (amongst other things) lithium is an uncompetitive inhibitor of myo-inositol monophosphate binding to inositol monophosphatase. This enzyme catalyzes the hydrolysis of myo-inositol monophosphate to myo-inositol, which is required in the phosphatidyl inositol cell signalling pathway, involved in many processes.

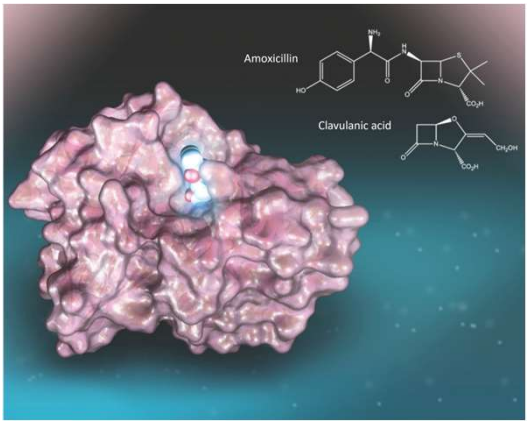
Noncompetitive inhibitor drugs include nevirapine and etravirine, used to manage HIV. They are noncompetitive non-nucleoside inhibitors (from the first and second generation, respectively) of the retroviral enzyme reverse transcriptase. Reverse transcriptase copies the viral RNA genome and makes a DNA copy, which is necessary for HIV replication. See https://pdb101.rcsb.org/motm/33.

Irreversible, Covalent Inhibitors: Examples



The multi-domain **penicillin-binding protein (PBP)** + the covalent inhibitor **penicillin**

Treats **bacterial infections** by blocking the formation of peptidoglycan



β-lactamase + the covalent inhibitor **clavulanic acid**

Ensures the efficacy of β-lactam antibiotics, like amoxicillin and penicillin, in **drug-resistant bacteria**

Figures: PDB-101: Learn: Flyers, Posters, & Calendars: Calendars: 2016 A Year in Protein-Drug Complexes and PDB-101: Global Health: Antimicrobial Resistance: undefined: Penicillin G and Structural Insights for β-Lactam Antibiotics

Irreversible, covalent inhibitors (also known as suicide or mechanism-based inhibitors) are relatively unreactive until they bind to the active site of a specific enzyme. There, they are chemically reactive and create an irreversible enzyme-inhibitor complex through a covalent bond. A covalent inhibitor generally resembles the substrate.

Covalent inhibitors, under conditions of $[E_0] >>> [I_0]$, resemble the kinetics of noncompetitive inhibitors. Both reduce the amount of active enzyme by a fixed amount.

Examples of irreversible, covalent inhibitor drugs include:

- The β-lactam antibiotic penicillin is a covalent inhibitor of penicillin-binding proteins, such as D-alanyl-D-alanine carboxypeptidase/transpeptidase, which are responsible for the cross-linking of peptidoglycan stem-peptides during cell wall synthesis. The β-lactam ring of penicillin forms a covalent bond with an active site serine. See https://pdb101.rcsb.org/motm/29.
- The β-lactam drug clavulanic acid is a covalent inhibitor of β-lactamase, an enzyme produced by bacteria that provides resistance to β-lactam antibiotics such as penicillin and amoxicillin. Its β-lactam ring forms a covalent bond with an active site serine. See https://pdb101.rcsb.org/global-health/antimicrobial-resistance/drugs/overcoming-resistance/inhibitors-of-beta-lactamases/clavulanic-acid/clavulanic-acid.
- The painkiller and blood thinner aspirin is a covalent inhibitor of cyclooxygenase (COX) 1 and 2. Both create prostaglandins which are used for basic housekeeping or signalling and inflammation. Aspirin is composed of an acetyl group attached to salicylic acid. Its acetyl group forms a covalent bond with an active site serine. See

https://pdb101.rcsb.org/motm/17.

- The medicine eflornithine (= difluoromethylornithine (DFMO)) is a covalent inhibitor of orthenine decarboxylase. This enzyme uses pyridoxal phosphate as a cofactor to catalyse the decarboxylation of ornithine to putrescine, a precursor for the polyamines spermine and spermidine — both essential in rapidly dividing cells. In *Trypanosoma* (the parasites that cause African sleeping sickness), ornithine decarboxylase is unusually long-lived, making it a good drug target. Eflornithine, a structural analogue of ornithine, forms a covalent bond with an active-site cysteine.

Properties of Drugs

High specificity:
- Targets only the intended enzyme or receptor.
- Reduces side effects caused by off-target binding.

Effective at low doses:
- High affinity (*a low* K_i) means smaller amounts are needed.
- Minimizes risk of toxicity.

Favourable pharmacokinetics:
- Drug remains stable in the body.
- Good solubility and distribution to reach the target site.
- Avoids accumulation that could lead to toxicity.

For an inhibitor to be a successful drug, it must have several key properties:
- It should be **highly specific**, meaning it binds only to the intended target and avoids off-target interactions that could cause side effects.
- It must be **effective at low doses**, indicating high potency and reducing the risk of **toxicity**.
- Additionally, it should have **favourable pharmacokinetics**, meaning it is stable in the body, has appropriate solubility and distribution to reach the target site, and does not accumulate in a way that could lead to toxicity.

Drug Discovery

- Serendipity (*e.g.* penicillin and sildenafil (Viagra))
- High-throughput screening
- Ligand-based or structure-based rational drug design

Drugs can be discovered through various approaches. Some are found by **chance**, such as **penicillin**, which was discovered when mould killed bacteria on a petri dish, and **sildenafil** (Viagra), which was originally developed for heart conditions but unexpectedly helped with erectile dysfunction.

Others are identified through **high-throughput screening**, where thousands of compounds are tested rapidly to find potential inhibitors.

More targeted approaches include **rational drug design**, which uses knowledge of molecular structures to create drugs that fit precisely into the active site of an enzyme. This can be **ligand-based**, using known molecules that bind to the target, or **structure-based**, using the 3D structure of the target to design complementary molecules.

Drug Discovery / Development Process







Research & Development	Pre-clinical research	Clinical Trials	Data Review & Approval	Post-market monitoring
2-4 years	**2-4 years**	**6-8 years**	**1-2 years**	**Ongoing**
Laboratory research. Target identification: Which pathway / molecule? Lead identification: Promising compounds	Laboratory and animal testing to assess efficacy and potential risk. If approved, moves to clinical trials.	Human trials to test effectiveness and safety.	Regulatory bodies review and examine data to decide on drug approval.	Ongoing safety monitoring.

The process of developing a new drug is long and complex, often taking **12-18 years**:

- It begins with **research and development**, which lasts around 2 to 4 years. During this phase, scientists conduct laboratory research to understand the disease and identify suitable **drug targets**, *i.e.* molecules or pathways that the drug will affect. Once a target is identified, researchers search for **lead compounds**, *i.e.* chemical structures that show potential for interacting with the target effectively.
- Following this, the drug enters the **pre-clinical research phase**, also lasting 2 to 4 years. Here, the drug is tested in laboratory settings and in animals to evaluate its efficacy and identify any major safety concerns. If the results are promising, the drug progresses to clinical trials.
- **Clinical trials** are conducted in humans and typically take 6 to 8 years. These trials are divided into phases: Phase I tests safety in healthy volunteers, Phase II assesses effectiveness and side effects in patients, and Phase III confirms the drug's efficacy in a larger population and compares it with existing treatments.
- Once clinical trials are complete, the data is submitted to regulatory authorities for review. This **data review and approval** stage takes about 1 to 2 years. Agencies such as the MHRA in the UK or the FDA in the US evaluate whether the drug is safe and effective enough to be approved for public use.
- Even after approval, the drug continues to be monitored through **post-market surveillance**. This ongoing process ensures that any long-term or rare side effects are identified and addressed, maintaining patient safety over time.

The approximate cost is around **$500 million to 1.5 billion**.

You Need to Be Able to…

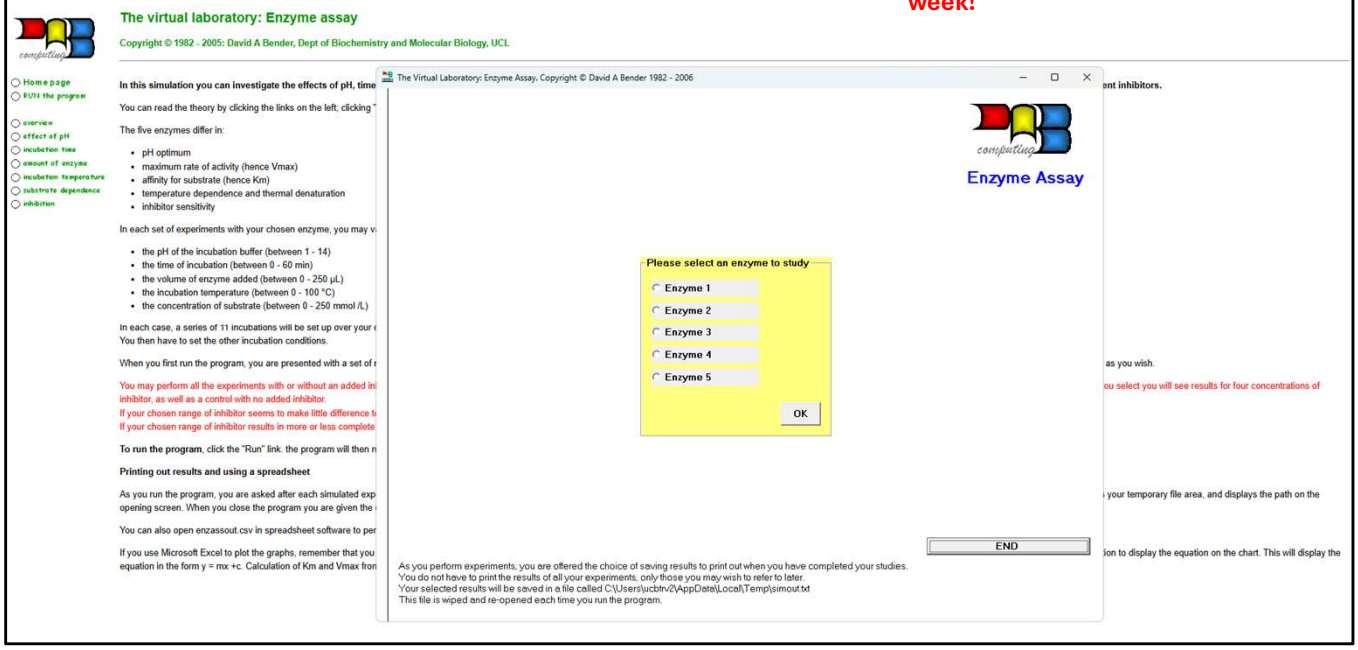
- ❑ provide examples of medicinal drugs that are enzyme inhibitors
- ❑ describe how covalent, competitive, uncompetitive, noncompetitive, and mixed inhibitors interact with the enzyme or enzyme-substrate complex
- ❑ compare and identify the effects of covalent, competitive, uncompetitive, noncompetitive, and mixed inhibition on enzyme activity
- ❑ calculate $K_{m,app}$, $V_{max,app}$, K_i, and K_i‘ for competitive, uncompetitive, noncompetitive, and mixed inhibitors

More Information?

- Lehninger Principles of Biochemistry; David L. Nelson, Michael M. Cox
  - 6.3 Enzyme Kinetics as an Approach to Understanding Mechanism
- Biochemistry; Jeremy M. Berg, Gregory J. Gatto Jr., Justin K. Jines, John L. Tymoczko, Lubert Stryer
  - 5.6 Enzymes Can Be Inhibited by Specific Molecules

Online Practical

Solutions released at the end of reading week!



Workshop

Deadline for your hot questions is 2 December 2026, 1 pm!

Workshop "Proteins and Enzymes"

We won't be able to go through every single question in our workshop. Instead, there is a Moodle "Hot Questions" activity where you can vote on the ones you'd most like us to cover. This way, we can focus on the topics that matter most to you.

Question 1. Which of the following is a sensible grouping based on their properties?

- ❑ Arg, Gly, Trp
- ❑ Asp, Ala, Phe
- ❑ Ala, Leu, Val
- ❑ Cys, Lys, Ile

Question 2. Which of the following pairs of amino acids would carry a positive charge on their sidechain at pH 5.0?

- ❑ Asparagine and Glutamine
- ❑ Histidine and Lysine
- ❑ Leucine and Glycine
- ❑ Aspartic acid and Glutamic acid

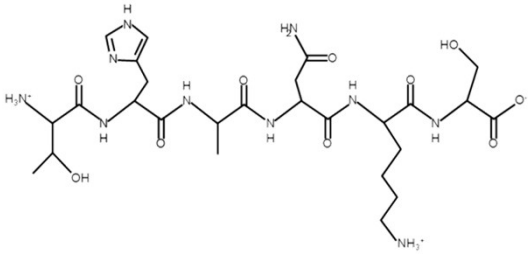
Question 3. Which amino acid has the lowest isoelectric point?

Use the easy-to-remember approximations for the pK_as from the α-carboxyl and α-amino groups and the sidechains.

- ❑ Gly
- ❑ Val
- ❑ Arg
- ❑ Asp

Question 4. If a lysine residue is near another basic sidechain in a protein's active site, the pK_a of its sidechain will most likely ...

T H A N K S



for your attention during these lectures!



https://www.youtube.com/watch?v=gAjR4_CbPpQ

*Work it **harder**, make it **better***
*Do it **faster**, makes us **stronger***
More than ever, hour after hour
Work is never over